M2795.007800
Guidance, Navigation, and Control System
for Reusable Launch Vehicles

Module 4: RLV Modeling & Closed-Loop Simulation



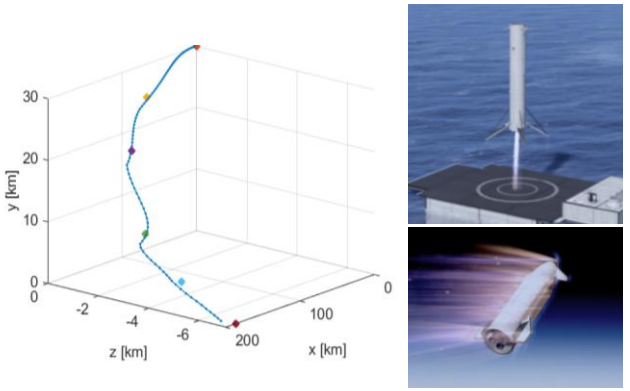
SpaceK-Big Project (1)

Development of Control Algorithms for Reusable Launch Vehicle

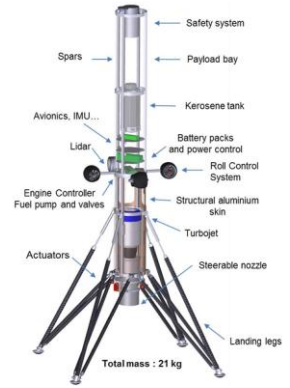
To support RLV development

- Develop high-fidelity software simulator and HILS platform
- Develop robust control algorithms optimized for each flight phase

1) High-fidelity validation framework

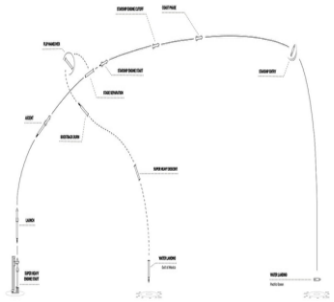


High-fidelity integrated simulator

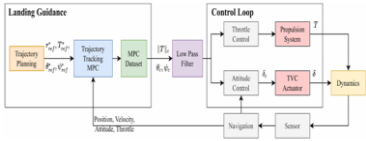


HILS platform

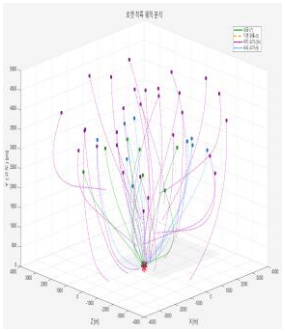
2) Core algorithms for RLV



ConOps



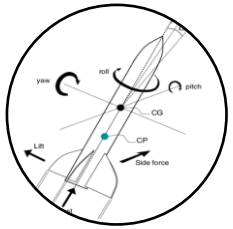
GNC algorithms



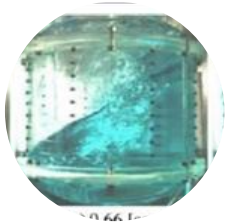
Simulation

SpaceK-Big Project (2)

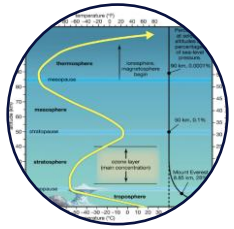
High-fidelity simulator



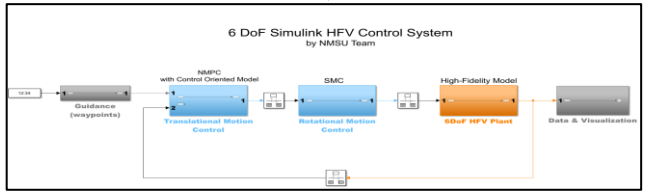
Dynamics



Sloshing

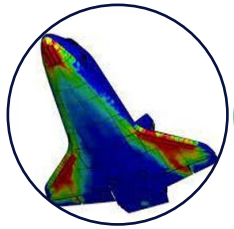


atm

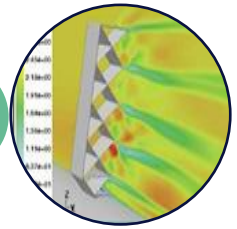


6-DoF Simulator

Control algorithm validation



CFD



Simulator integration



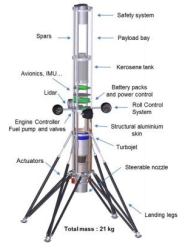
Grid fin,
TVC control



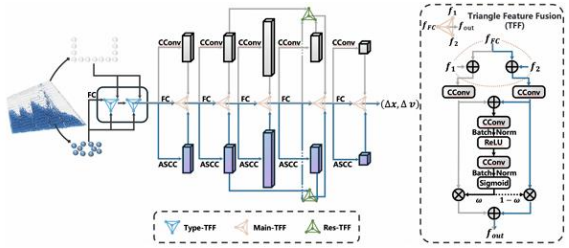
Control
surface

HILS platform





HILS platform for validation



Neural network-based
algorithm adaptation

Goal Outcomes

- Mission scenario-based high-fidelity simulator
- Robust control algorithms for precise landing
- Hardware platform for landing phase validation

Contents



- Part 1
  - Introduction
  - Dynamics
  - Aerodynamics
  - Closed-loop Simulation

- Part 2
  - Guidance Model and Problem Formulation
  - Trajectory Optimization and Online SCP Simulation
  - SRML RLV Guidance Research

Speaker Introduction



ABOUT ME

Junesuk Kim

Master student

Email: junesuk612@snu.ac.kr

Space Mobility & Robotics Laboratory

(Advisor: Dr. Hyeongjun Park)

Research interest:

Control allocation on aerospace application:

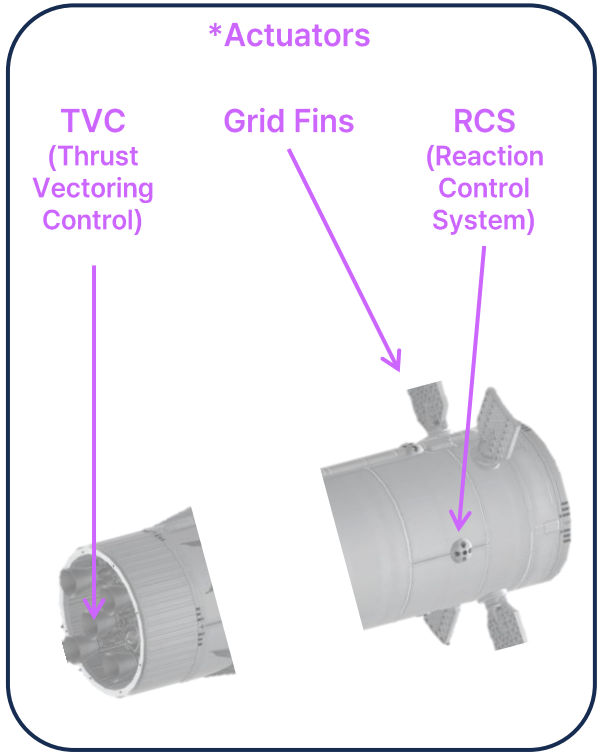
Null-space drag modulation with RLV's grid-fin

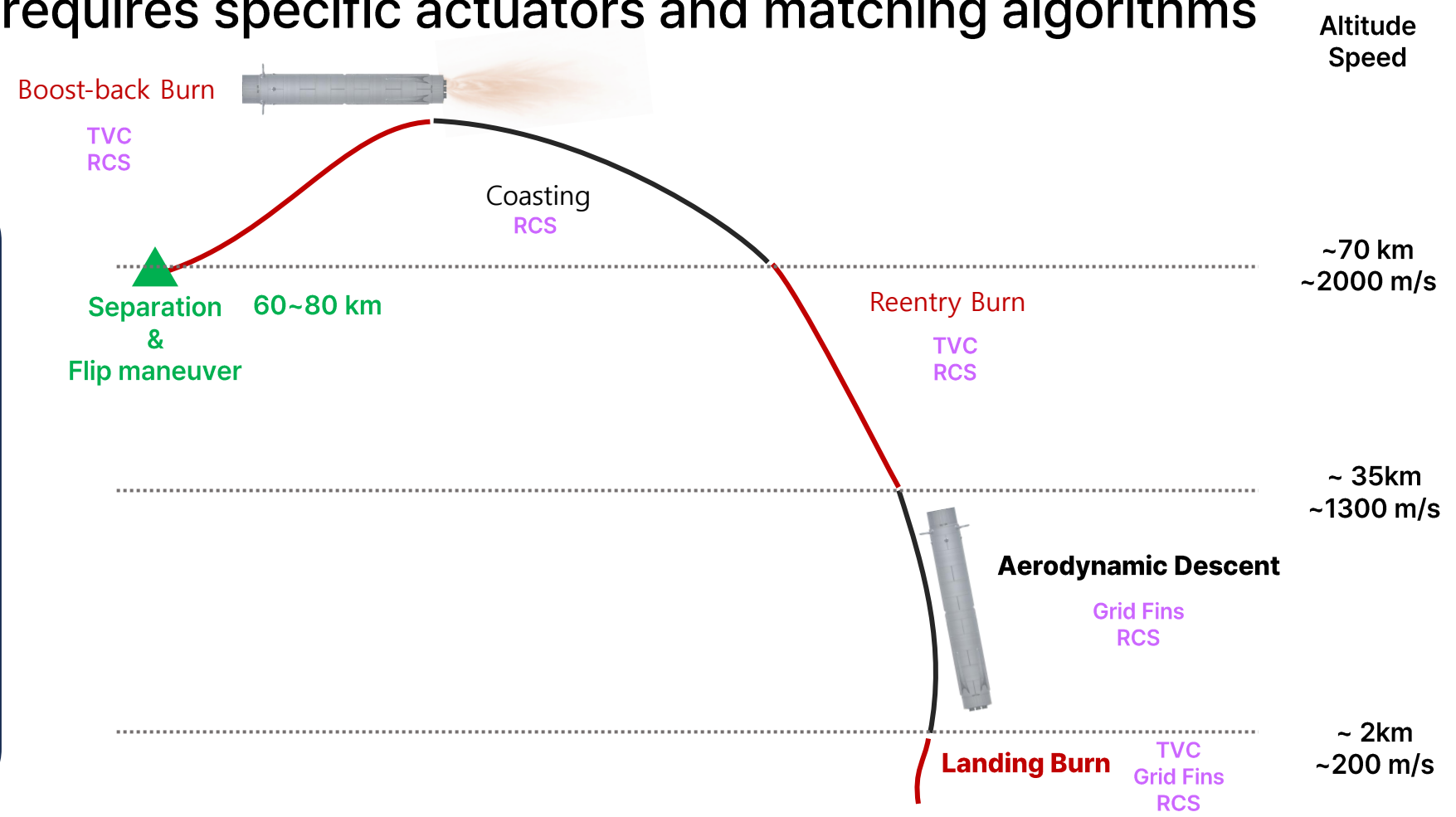
- Conference proceedings
  - **J. Kim**, K. Jang, and H. Park, "Null-Space Drag Modulation with Grid Fins: A Load-Decoupled Downrange Control Channel for Reusable Launch Vehicle Aerodynamic Descent," 77th International Astronautical Congress (IAC 2026), Antalya, Türkiye, Oct. 2026
  - K. Jang, **J. Kim**, and H. Park, "Robust Multi-Phase Descent Guidance for a Reusable Launch Vehicle," 77th International Astronautical Congress (IAC 2026), Antalya, Türkiye, Oct. 2026
  - **J. Kim**, and H. Park, " Null-Space Reinforcement Learning for Trajectory Optimization of Free-Floating Space Manipulators under Inertia Changes," 23rd IFAC World Congress, 2026.
  - **김준석**, 장강묵, 박형준, "강화학습 기반 적응형 가중치 튜닝을 이용한 재사용 발사체 최적제어 연구," 한국항공우주학회 춘계학술대회, 제주, 2026년 4월.
  - **김준석**, 박형준, "QP-DDPG 기반 무인 우주 정거장 로봇팔 반복 궤적 자율 운용," 한국항공우주학회 추계학술대회, 고성, 2025년 11월.

Introduction

Introduction

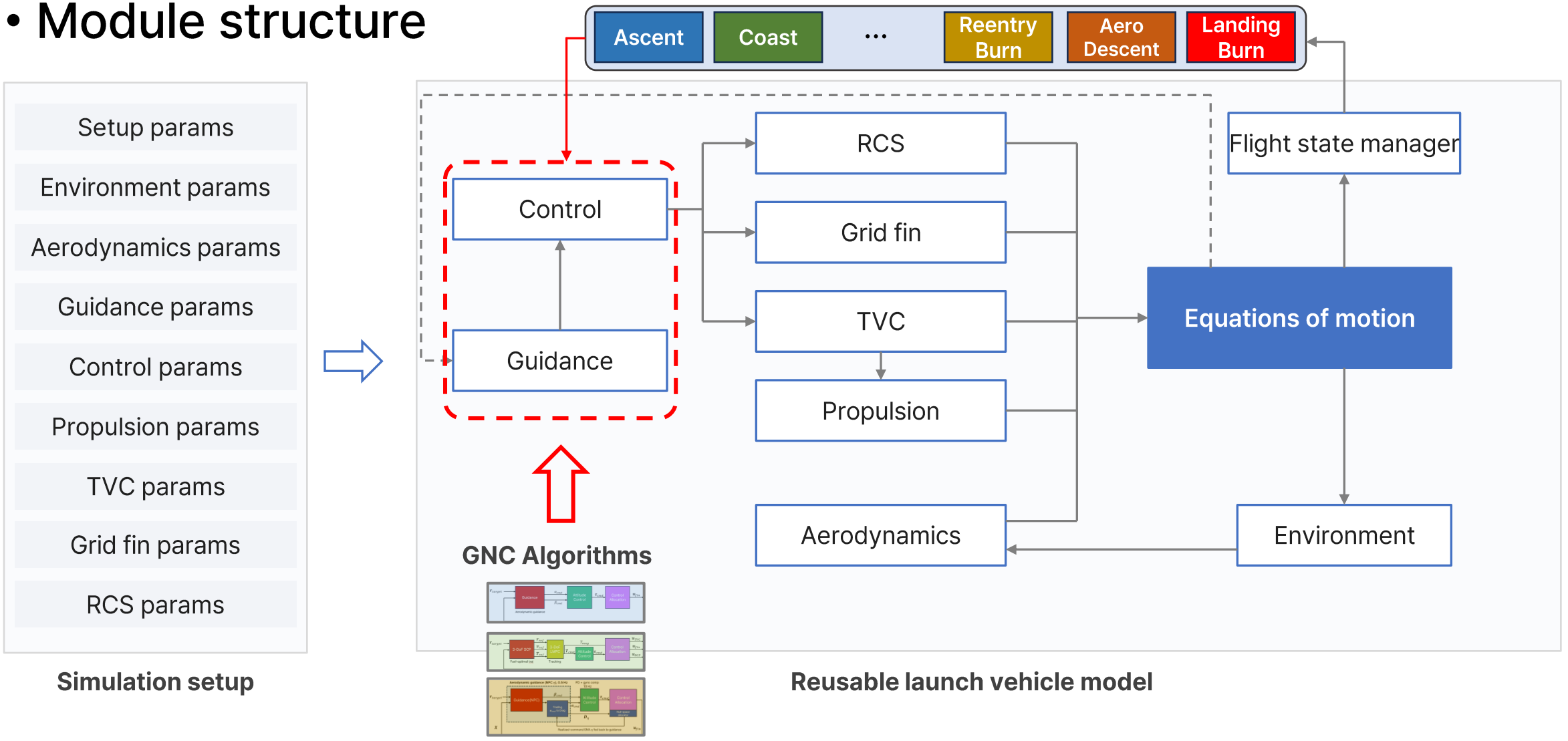
- Landing phases
  - Each phase requires specific actuators and matching algorithms





Architecture

- Module structure



Dynamics

Dynamics

- 6DoF Dynamics

$$\mathbf{x} = \begin{bmatrix} \mathbf{r}_I \\ \mathbf{v}_B \\ \mathbf{q}_{B/I} \\ \boldsymbol{\omega}_B \\ m \end{bmatrix} \in \mathbb{R}^{14\times 1} \quad \mathbf{u} = \begin{bmatrix} \mathbf{u}_{TVC} \\ \mathbf{u}_{fin} \\ \mathbf{u}_{RCS} \end{bmatrix} \in \mathbb{R}^{(3+4+n)\times 1}$$

$$\mathbf{F}_{total} = \mathbf{F}_g + \mathbf{F}_{aero} + \mathbf{F}_{TVC} + \mathbf{F}_{RCS}$$

$$m(\dot{\mathbf{v}}_b + \boldsymbol{\omega}_b \times \mathbf{v}_b) = \mathbf{F}_{total}$$

$$\mathbf{M}_{total} = \mathbf{M}_{aero} + \mathbf{M}_{TVC} + \mathbf{M}_{RCS} + \mathbf{M}_{damping}$$

$$\mathbf{I}\dot{\boldsymbol{\omega}}_b + \boldsymbol{\omega}_b \times (\mathbf{I}\boldsymbol{\omega}_b) = \mathbf{M}_{total}$$

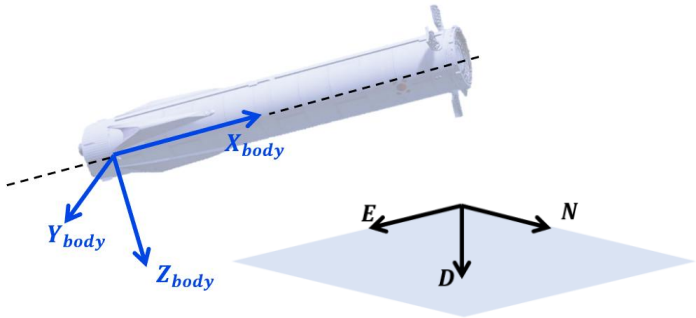
$\mathbf{r}_I = [x\ y\ z]^T_{NED}$

$\mathbf{v}_B = [u\ v\ w]^T_{Body}$

$\mathbf{q}_{B/I} = [q_0\ q_1\ q_2\ q_3]^T_{Body}$

$\boldsymbol{\omega}_B = [p\ q\ r]^T_{Body}$

$m \in \mathbb{R}$



Dynamics - Translation

- Forces

$$\mathbf{F}_{total} = \mathbf{F}_g + \mathbf{F}_{aero} + \mathbf{F}_{TVC} + \mathbf{F}_{RCS}$$

$$m(\dot{\mathbf{v}}_b + \boldsymbol{\omega}_b \times \mathbf{v}_b) = \mathbf{F}_{total}$$

$$\dot{\mathbf{v}}_b = \frac{\mathbf{F}_{total}}{m} - \boldsymbol{\omega}_b \times \mathbf{v}_b$$

$\mathbf{F}_g$ **Gravity**

g = μ/(Re + h)², acting straight down in NED; rotated into the body frame with the attitude quaternion

$\mathbf{F}_{aero}$ **Aerodynamic force**

Uses air density and speed of sound from the ISA model, wind-relative velocity, and the four grid-fin deflections

$\mathbf{F}_{TVC}$ **Thrust (TVC)**

T = throttle × T_vac along the gimbal direction (δy, δz), cut off at dry mass; the engine's offset from the CG also produces a moment

$\mathbf{F}_{RCS}$ **Reaction control system (RCS)**

Small thrusters around the vehicle that fire in pairs for attitude torque with little net force; used where grid fins and TVC lack authority

Dynamics - Rotation

- Moments

$$\mathbf{M}_{total} = \mathbf{M}_{aero} + \mathbf{M}_{TVC} + \mathbf{M}_{RCS}$$

$$\mathbf{I}\dot{\boldsymbol{\omega}}_b + \boldsymbol{\omega}_b \times (\mathbf{I}\boldsymbol{\omega}_b) = \mathbf{M}_{total}$$

$$\dot{\boldsymbol{\omega}}_b = \mathbf{I}^{-1}\left(\mathbf{M}_{total} - \boldsymbol{\omega}_b \times (\mathbf{I}\boldsymbol{\omega}_b)\right)$$

$\mathbf{M}_{aero}$ **Aerodynamic moment**

Grid-fin and body aero forces acting about the current CG; x_cg and inertia I are recomputed from the remaining mass each step

$\mathbf{M}_{TVC}$ **Thrust (TVC) moment**

M = r_eng × F_TVC, with r_eng = [−x_cg, 0, 0]; gimbal angles (δy, δz) produce pitch and yaw torque

$\mathbf{M}_{RCS}$ **Reaction control (RCS) moment**

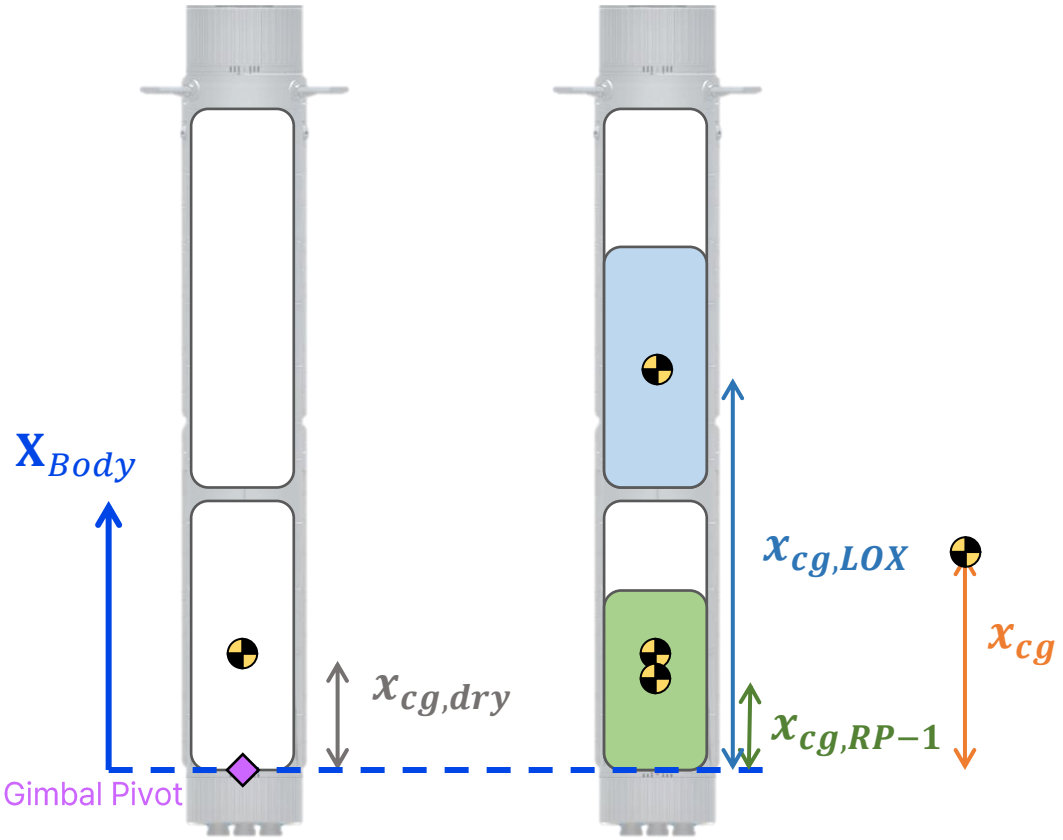
Paired thrusters fire on opposite sides to produce a pure torque, mainly for roll control and attitude hold at low dynamic pressure

- Mass

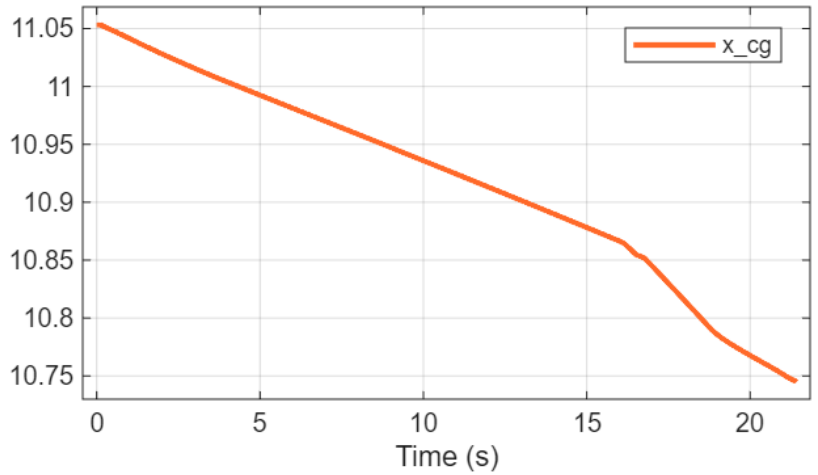
$$\dot{m} = -\frac{T}{I_{sp}\, g_0}$$

Inertia Update

- Time-varying geometry: Inertia
  - Dual tank model



Time – CoG position plot while PDG



O/F Ratio: 2.56(Merlin 1D)

$$m_{rp-1} = \frac{m_{prop}}{OF_{Ratio} + 1}$$

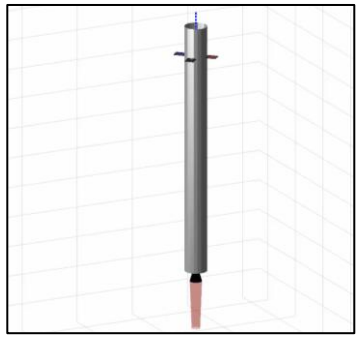
$$x_{cg} = \frac{(m_{dry} \cdot x_{cg_dry} + m_{lox} \cdot x_{cg_lox} + m_{rp1} \cdot x_{cg_rp1})}{m}$$

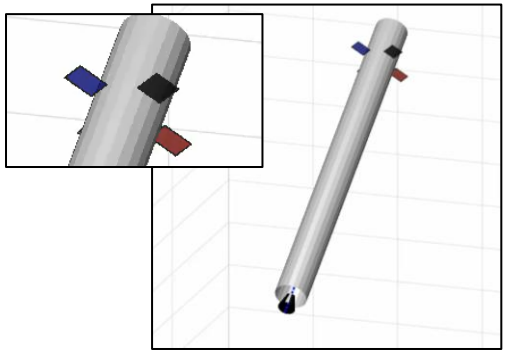
$$I_{total} = \sum_{i=1}^{n} (I_{\text{base},i} + m_i \cdot d_i^2)$$

Actuators

- Actuators overview
  - TVC – thrust magnitude + 2-axis gimbal

$$T = \tau \cdot (T_{vac} - P_{amb}A_e) \qquad \hat{n} = \begin{bmatrix} \cos\delta_y \cos\delta_z \\ \sin\delta_z \\ \sin\delta_y \end{bmatrix} \qquad \mathbf{F}_b = T \cdot \hat{n} = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} \qquad \mathbf{M}_b = \mathbf{r}_{cg} \times \mathbf{F}_b$$

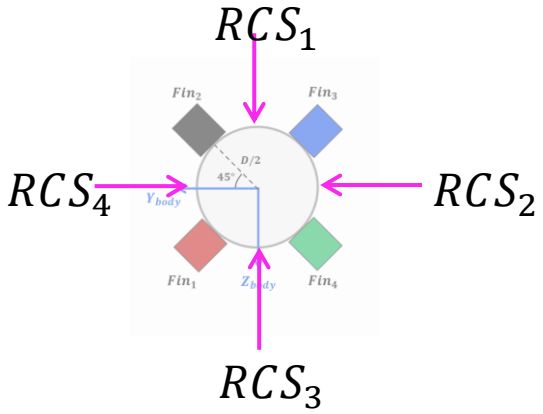


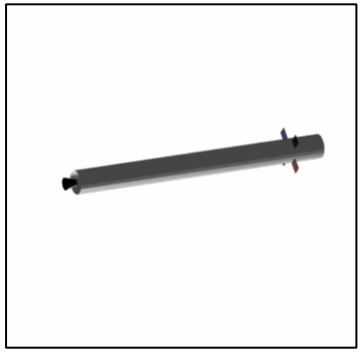


  - Grid fins – 4 independent deflection angles

  - RCS – 4 on/off thrusters

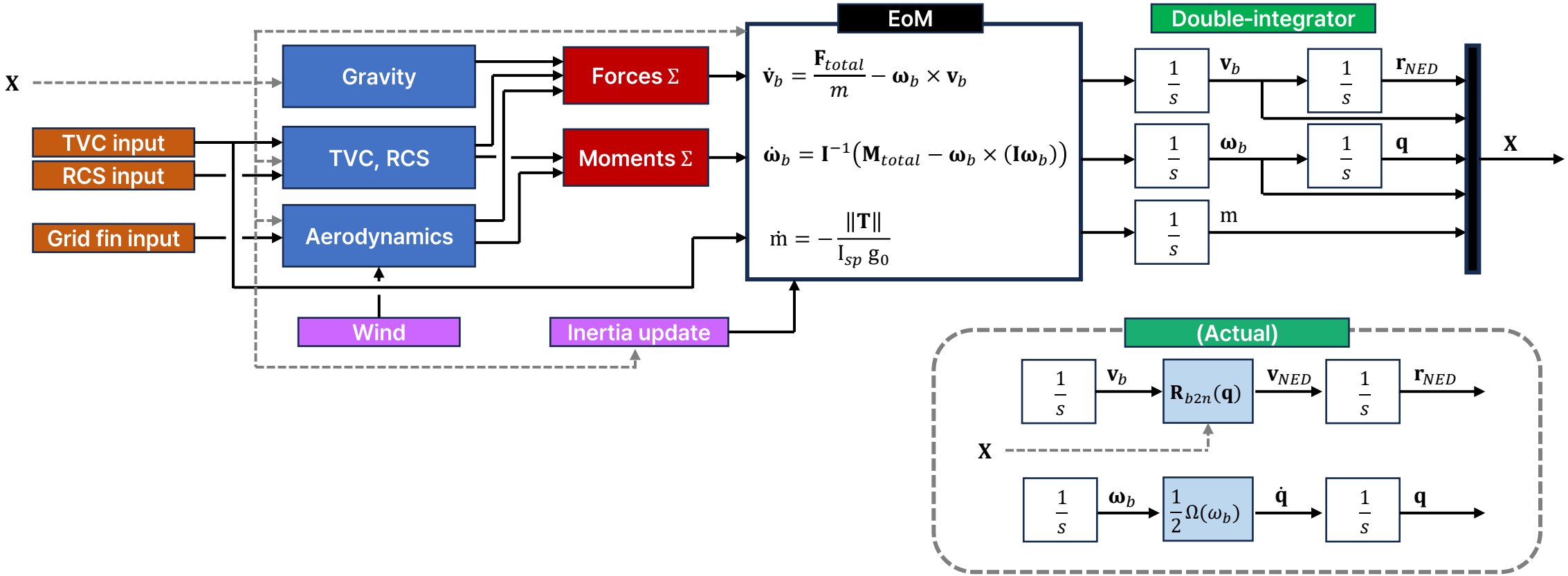
$$\mathbf{F}_{RCS,b} = \begin{bmatrix} 0 \\ (u_{y+} - u_{y-}) \cdot F_{max} \\ (u_{z+} - u_{z-}) \cdot F_{max} \end{bmatrix} \qquad \mathbf{M}_b = \mathbf{r}_{eng} \times \mathbf{F}_b$$





Dynamics

- Integration
  - 6-DoF rigid-body EoM in the body frame
  - Position and attitude via kinematics
  - Variable mass



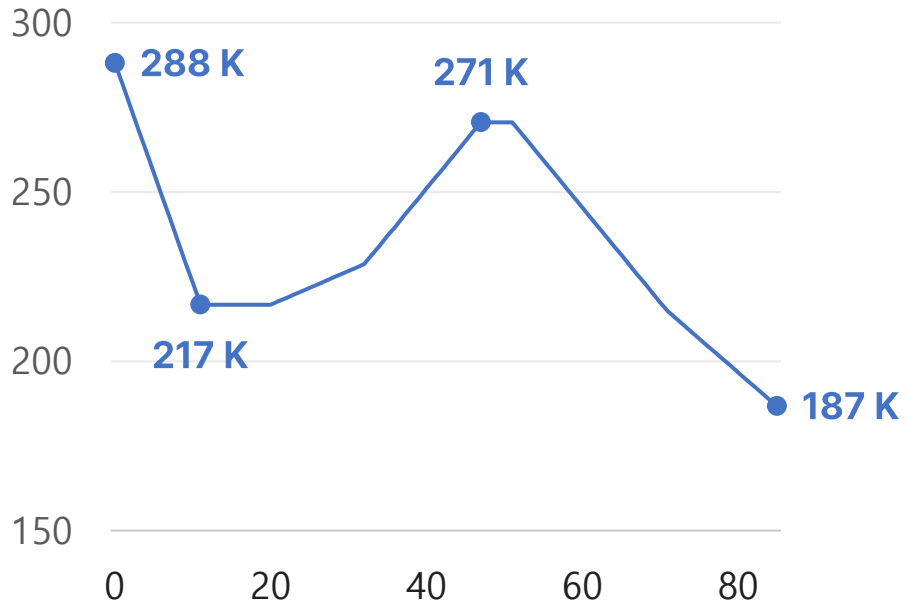
Aerodynamics

$\mathbf{F}_{aero}$ & $\mathbf{M}_{aero}$

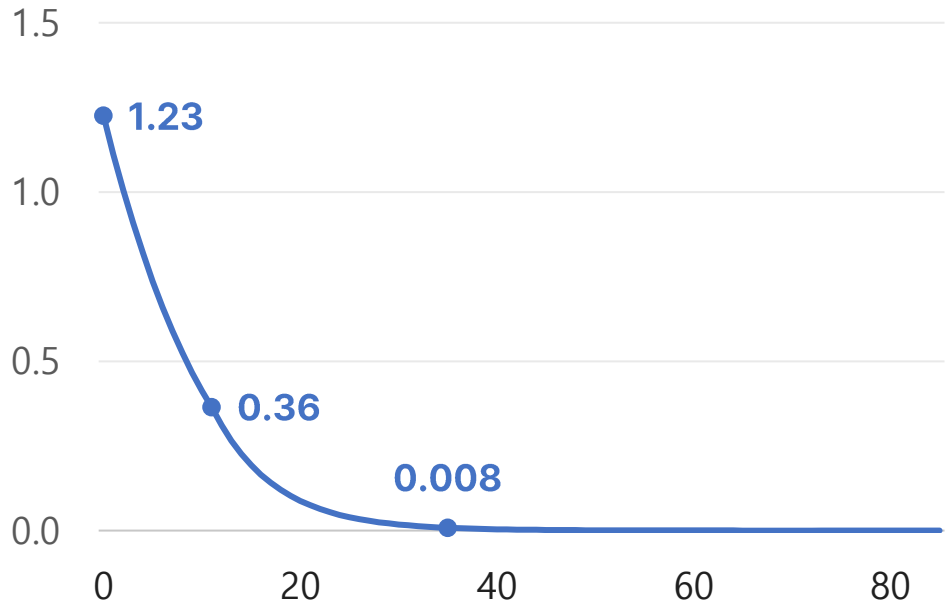
Atmosphere Model

- Atmosphere model: extended ISA
  - 7 layers up to 84.9 km, each with a linear temperature lapse rate
  - Gives density ρ = P/RT and speed of sound a = √(γRT) for the aero model

Temperature vs altitude, K



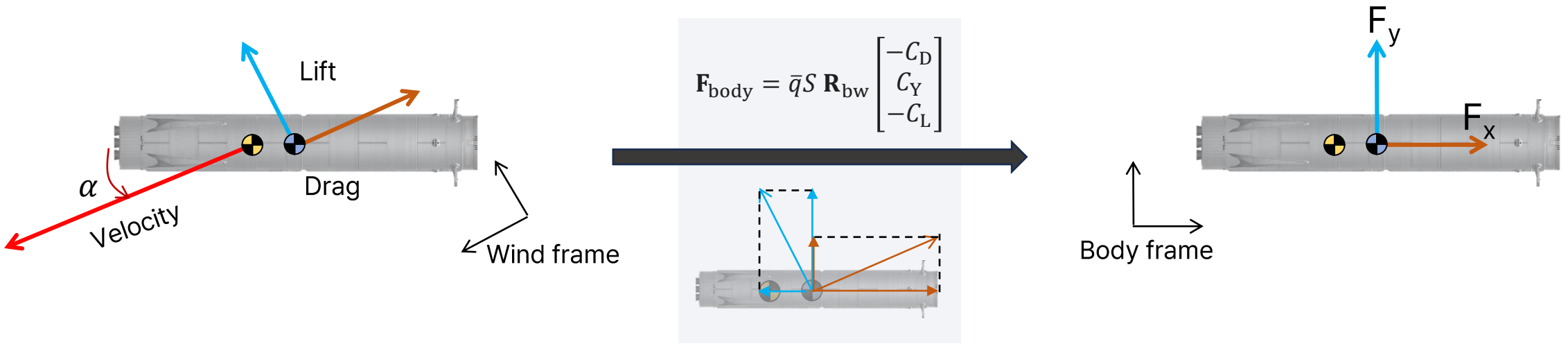
Density vs altitude, kg/m³



x-axis: altitude, km. Source: U.S. Standard Atmosphere, 1976

Aerodynamics

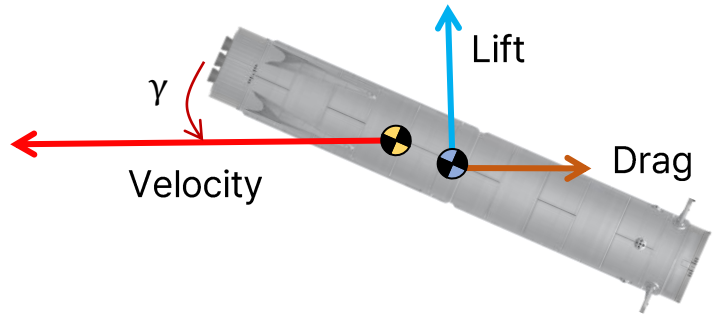
- Wind Frame
  - Lift and drag are defined in the wind frame (x_w along airspeed)
  - The equations of motion use the body frame, so rotate them by $R_{bw}(\alpha, \beta)$ to get F_x, F_y, F_z



  - The body force acts at the CP; its offset from the CG creates a moment that sets the vehicle's static stability.

Aerodynamics – Body

- Body aero
  - The body generates lift and drag that depend on the angle of attack, α (AoA)
  - Axisymmetric, so the same relation holds for side force and sideslip angle, β (SSA)
  - AoA and SSA are combined into a total AoA,γ



1 Flow angle

$\gamma = \tan^{-1}\left(\sqrt{v^2 + w^2}/|u|\right)$

Total AoA from body-frame airspeed

2 Coefficients ***

$C_{\mathrm{D}} = C_{\mathrm{D0}} + K_2\gamma^2 + K_4\gamma^4$

$C_{\mathrm{N}} = C_{\mathrm{N1}}\gamma + C_{\mathrm{N3}}\gamma^3$

Drag: even, 4th order in γ.
Normal: odd, 3rd order

3 Body force

$\mathbf{F}_{\mathrm{body}} = \bar{q}S\,\mathbf{R}_{\mathrm{bw}}\begin{bmatrix} -C_{\mathrm{D}} \\ C_{\mathrm{Y}} \\ -C_{\mathrm{L}} \end{bmatrix}$

4 Body moment

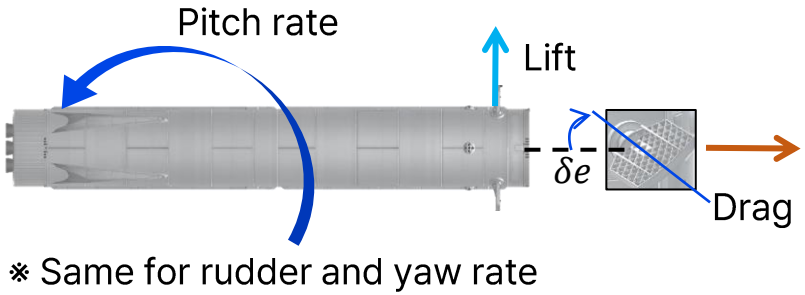
$\mathbf{M}_{\mathrm{body}} = \mathbf{r}_{\mathrm{offset}} \times \mathbf{F}_{\mathrm{body}} + \mathbf{M}_{\mathrm{damp}}$

damping ∝ p, q, r

$\mathbf{F}_{\mathrm{aero}} = \mathbf{F}_{\mathrm{body}} + \mathbf{F}_{\mathrm{fin}}$, $\mathbf{M}_{\mathrm{aero}} = \mathbf{M}_{\mathrm{body}} + \mathbf{M}_{\mathrm{fin}}$ Grid fins add only the deflection increment

Aerodynamics – Control Surface

- Control surface
  - In aero descent, AoA is varied to steer the trajectory
  - AoA is set by actuated control surfaces
  - Elevator for pitch, rudder for yaw, aileron for roll



1 Actuator
δ_e
Command deflects the grid fins

2 Fin force
$L_{\text{fin}} \propto \bar{q} S_{\text{fin}} C_{L\alpha} \delta$
Lift from the fin deflection

3 Control moment
$\mathbf{M}_{\text{fin}} = \mathbf{r}_{\text{offset}} \times \mathbf{F}_{\text{fin}}$
Fin offset from CG pitches the vehicle

4 Body AoA
$q \rightarrow \alpha \uparrow$
Vehicle rotates; body AoA builds

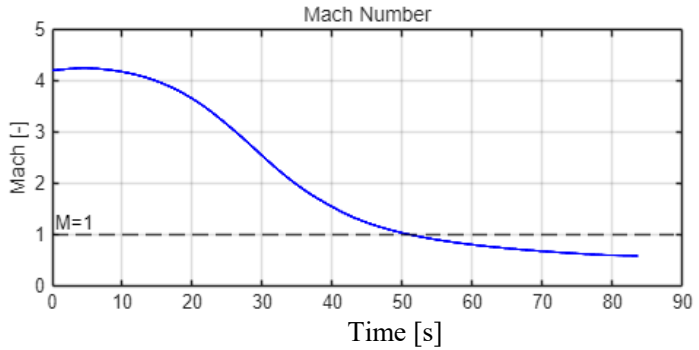
5 Body lift
$C_{\text{N}} = s_{\text{N}}(M) f_{\text{N}}(\gamma)$
Body lift bends the trajectory

  - Fin forces also add to the vehicle's net force

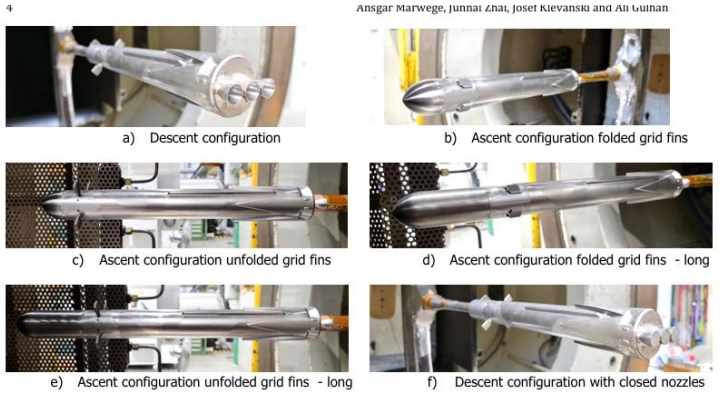
$$\mathbf{F}_{\text{aero}} = \mathbf{F}_{\text{body}} + \mathbf{F}_{\text{fin}}, \quad \mathbf{M}_{\text{aero}} = \mathbf{M}_{\text{body}} + \mathbf{M}_{\text{fin}}$$

Aerodynamics

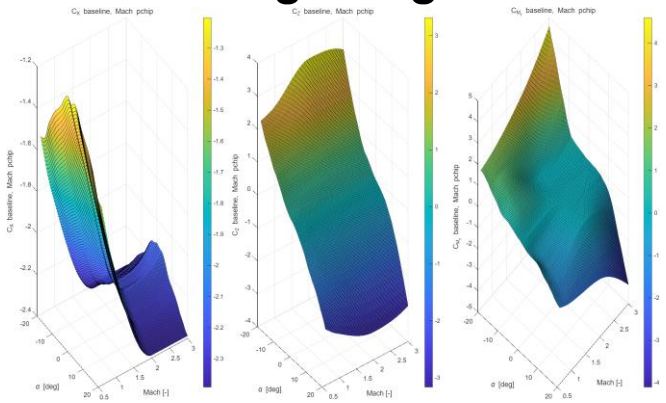
- Aero coefficient model
  - Coefficients are nonlinear in both AoA and Mach
  - Digitized from wind tunnel data in the literature
  - Fitted with polynomials for use in the GNC algorithm
  - Mach range: descent from 35 km at 1,300 m/s →



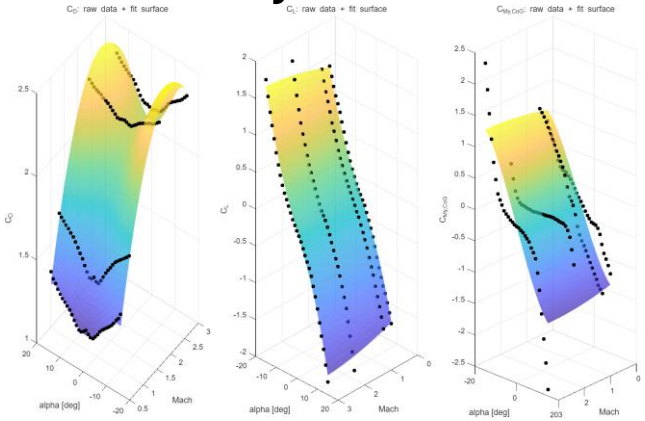
Wind tunnel data



Digitizing



Polynomial

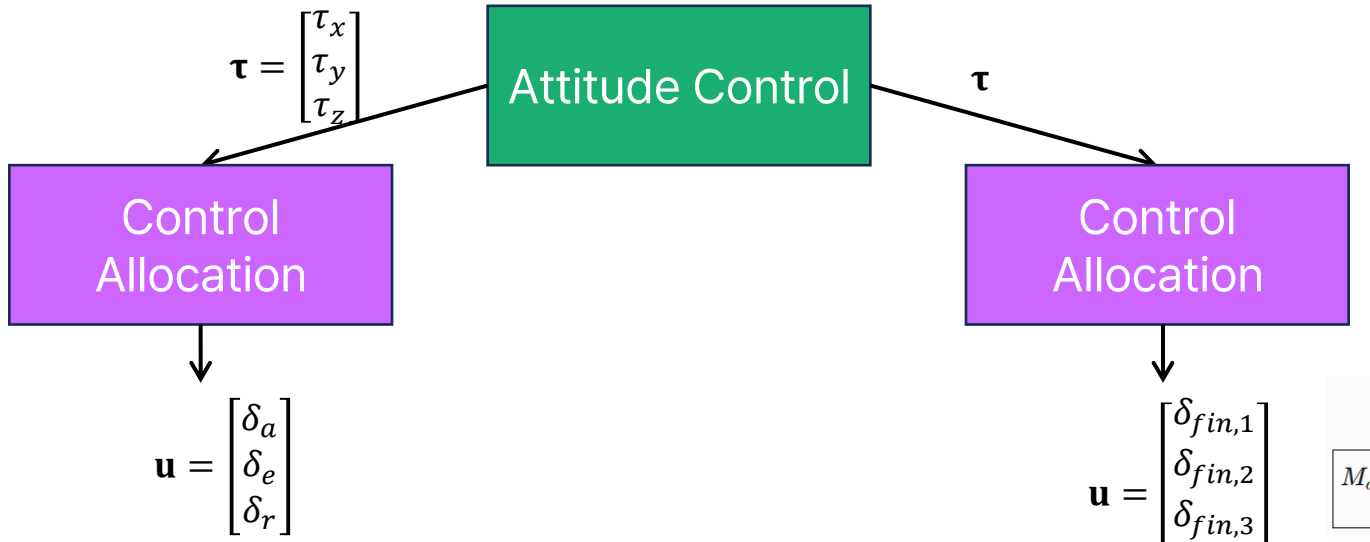


Marwege, Ansgar, et al. "Aerodynamic Behavior of a Descending Launcher First Stage Demonstrator with Grid Fins in the Aerodynamic Phase." (2026).

Aero Model: 3-channel vs 4-fin

- Virtual input vs explicit allocation
  - Attitude control outputs a torque command τ; allocation turns it into fin deflections
  - Virtual input: τ → (δ_a, δ_e, δ_r) mixed to 4 fins; simple, ignores fin coupling
  - Explicit allocation: τ → each $\delta_{fin,i}$ from the fin model; handles coupling, limits
  - The right choice depends on what you want to study

$$L = (C_{L0} + C_L^{\alpha}\alpha + C_L^{\delta_e}\delta_e)q_0 S_r$$
$$D = (C_{D0} + C_D^{\alpha}\alpha + C_D^{\alpha^2}\alpha^2)q_0 S_r$$
$$Y = (C_{L0} - C_L^{\alpha}\beta - C_L^{\delta_e}\delta_r)q_0 S_r$$
$$M_x = \left(\frac{m_x^p L_r}{V} + m_x^{\beta}\beta + m_x^{\delta_a}\delta_a\right)q_0 S_r L_r$$
$$M_y = \left(\frac{m_y^q L_r}{V} + m_y^{\alpha}\alpha + m_y^{\delta_e}\delta_e\right)q_0 S_r L_r$$
$$M_z = \left(\frac{m_z^r L_r}{V} + m_z^{\beta}\beta + m_z^{\delta_r}\delta_r\right)q_0 S_r L_r$$



<Virtual Input>

<Explicit allocation>

$$F_{aero} = F_{body}(\gamma, M) + \Delta F_{gridfin}(u_g, V_{local}, M)$$
$$M_{aero} = r_{cp} \times F_{body} + M_{damping} + \sum_i r_{fin,i} \times \Delta F_{fin,i}$$

Zhang, Liang, et al. "Adaptive fault-tolerant control for a VTVL reusable launch vehicle." Acta Astronautica 159 (2019): 362-370.

4-fin Modeling

- Grid fins – geometry & local frames
  - Each fin is modeled in its own local frame
  - Freestream in each local frame gives the fin AoA
  - Fin forces sum into vehicle force and moment

- Grid-fin aerodynamics
  - Fin lift is linear in AoA; fin drag is quadratic

1 Local fin AoA

$\mathbf{V}_{\mathrm{loc,i}} = \mathbf{V}_{\mathrm{air}} + \boldsymbol{\omega} \times \mathbf{r}_i$

$\alpha_i = \delta_i - \varphi_i$

φ_i = local flow angle at fin i

Body rates shift the local flow

2 Fin force increment

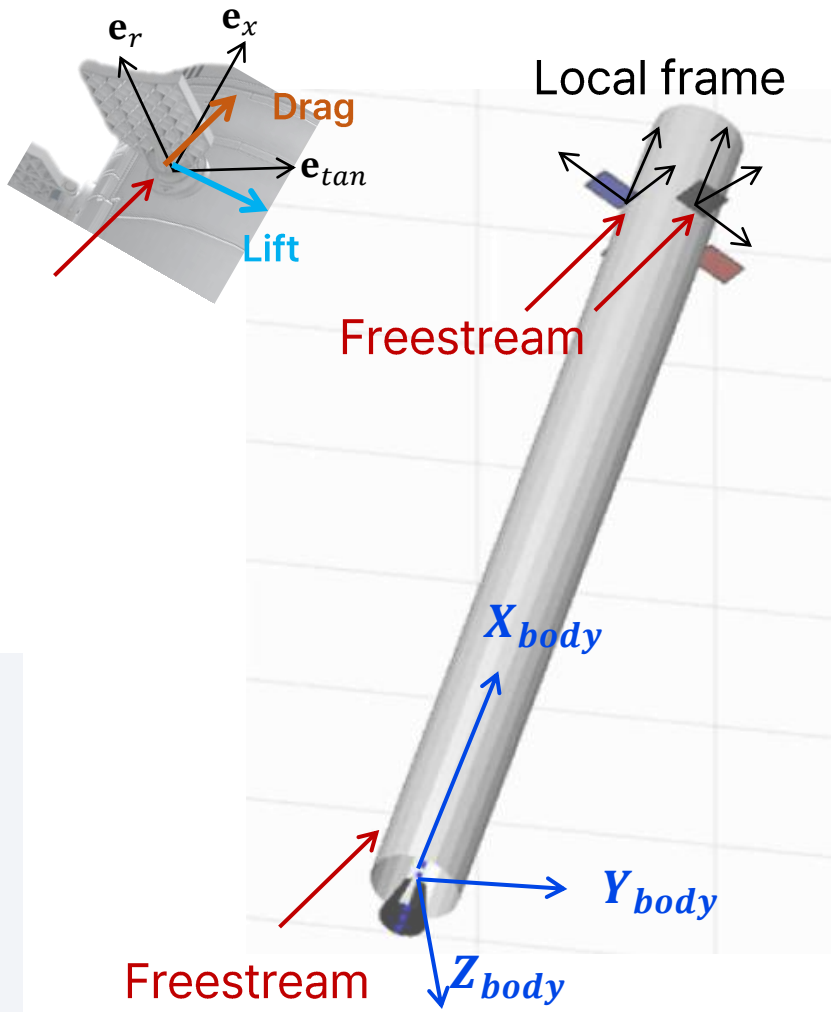
$L_i \propto C_L(\alpha_i - \alpha_{0,i})$

$D_i \propto C_D(\alpha_i^2 - \alpha_{0,i}^2)$

$\mathbf{F}_{\mathrm{fin}} = \sum_i (L_i \mathbf{e}_{\tan,i} - D_i \hat{\mathbf{v}}_i)$

Scaled by dynamic pressure × fin area

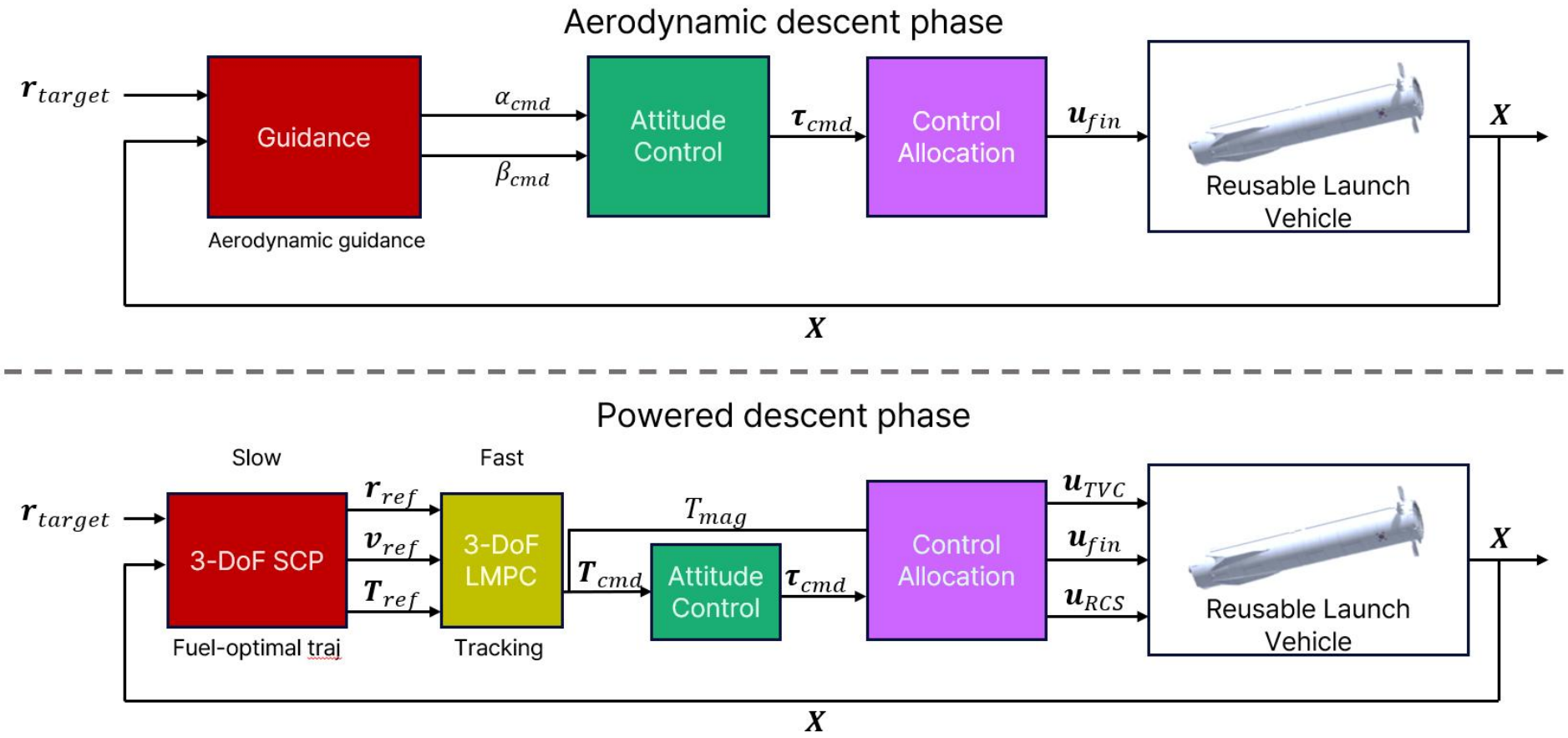
α_0 = neutral fin, already in body aero



Closed-loop Simulation

Simulation Loops

- From guidance to actuator
  - Each flight phase runs its own GNC loop

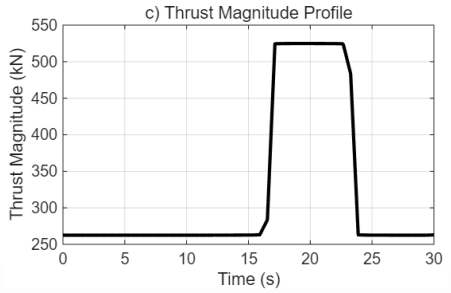


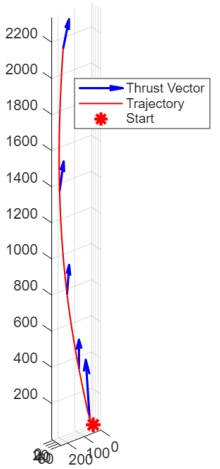
Jung, Ki-Wook, et al. "Optimal landing guidance of reusable launch vehicles with dual-convex optimization." International Journal of Control, Automation and Systems 23.8 (2025): 2198-2218.

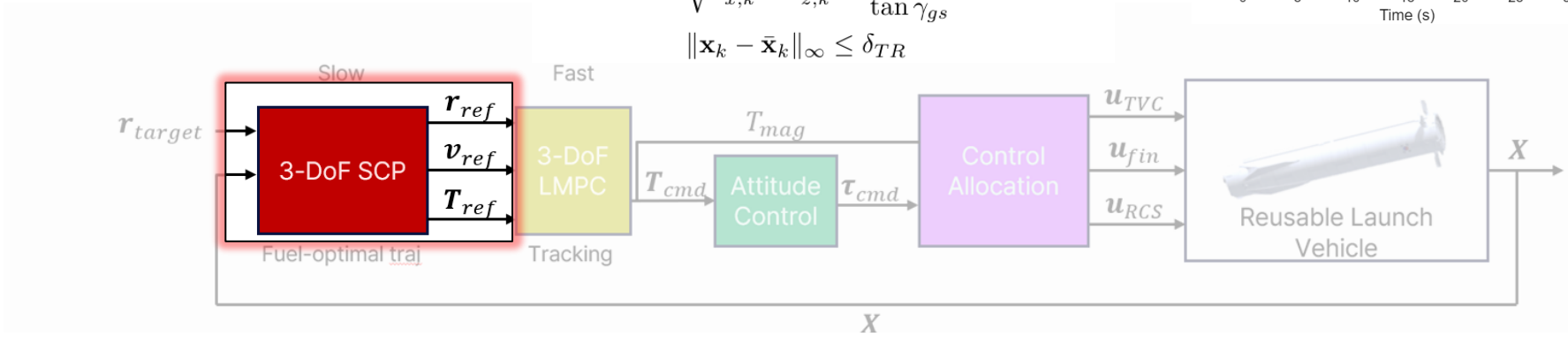
Guidance – PDG

- Guidance
  - Plans a fuel-optimal trajectory and thrust profile to the target
  - SCP on a 3-DoF point-mass model for real time
  - Enforces thrust and glideslope constraints
  - Runs at a slow update rate

$$
\begin{aligned}
\underset{\mathbf{x},\mathbf{u},\sigma}{\text{minimize}} \quad & -m(t_f) \quad \text{(Fuel Optimization)} \\
\text{subject to} \quad & \text{Linearized Dynamics (Eq. 8)} \\
& \mathbf{x}_0 = \mathbf{x}_{init}, \quad \mathbf{x}_N = \mathbf{x}_{target} \\
& \|\mathbf{u}_k\|_2 \le \sigma_k, \quad T_{min} \le \sigma_k \le T_{max} \\
& \sqrt{r_{x,k}^2 + r_{z,k}^2} \le \frac{r_{y,k}}{\tan \gamma_{gs}} \\
& \|\mathbf{x}_k - \bar{\mathbf{x}}_k\|_\infty \le \delta_{TR}
\end{aligned}
$$







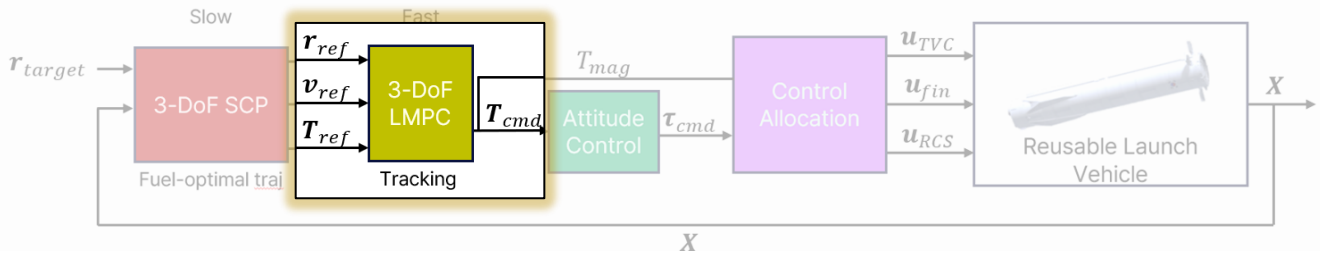
Ma et al. "Picard iteration-based convexification for fuel-optimal rocket descent inside atmosphere." Journal of Guidance, Control, and Dynamics 46.2 (2023): 343-349.

Control – PDG

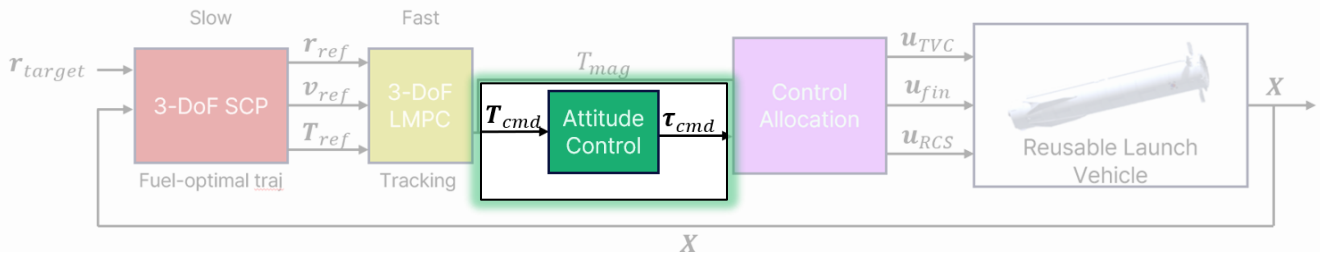
- Controller - Tracking
  - MPC tracks the SCP trajectory
  - The SCP solution is too coarse in time to use directly
  - Outputs the 3D thrust vector needed to track it (Fast loop)

$$\mathbf{x}_k = [r_k, v_k]^T \in \mathbb{R}^6 \quad \mathbf{u}_k \in \mathbb{R}^3 \quad J = \sum_{i=1}^{N_h} (\mathbf{x}_i - \mathbf{x}_{ref,i})^T \mathbf{Q} (\mathbf{x}_i - \mathbf{x}_{ref,i}) + \sum_{i=0}^{N_h - 1} \mathbf{u}_i^T \mathbf{R} \mathbf{u}_i$$

$$\mathbf{Q} \in \mathbb{R}^{6\times6} \quad \mathbf{R} \in \mathbb{R}^{3\times3}$$



- Controller – Attitude
  - Thrust direction sets the target RLV attitude
  - Outputs the torque command to track it



Control Allocation - PDG

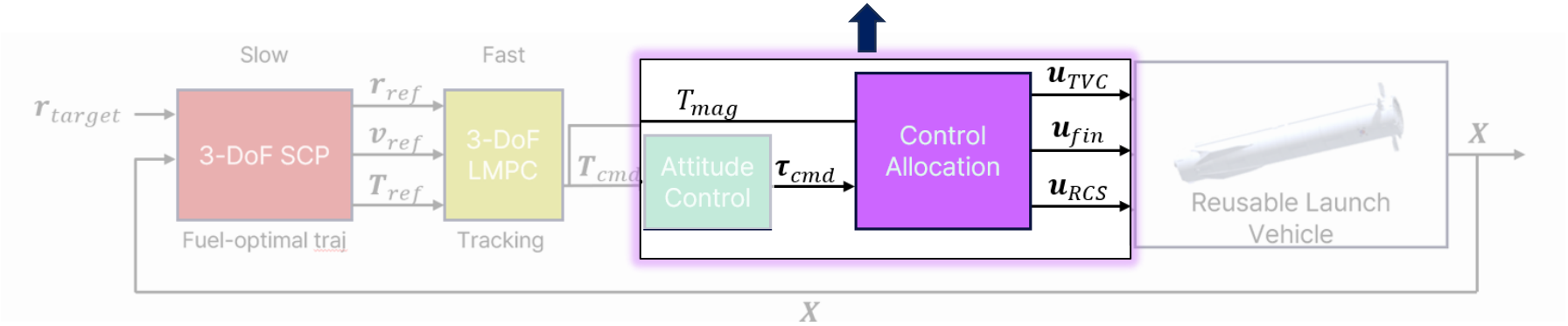
- Control allocation
  - Splits the torque command across actuators
  - Pseudo-inverse or QP gives optimal inputs

$$\boldsymbol{\tau} \in \mathbb{R}^3 \qquad \boldsymbol{u} \in \mathbb{R}^{10} \qquad \boldsymbol{B} \in \mathbb{R}^{3\times 10}$$

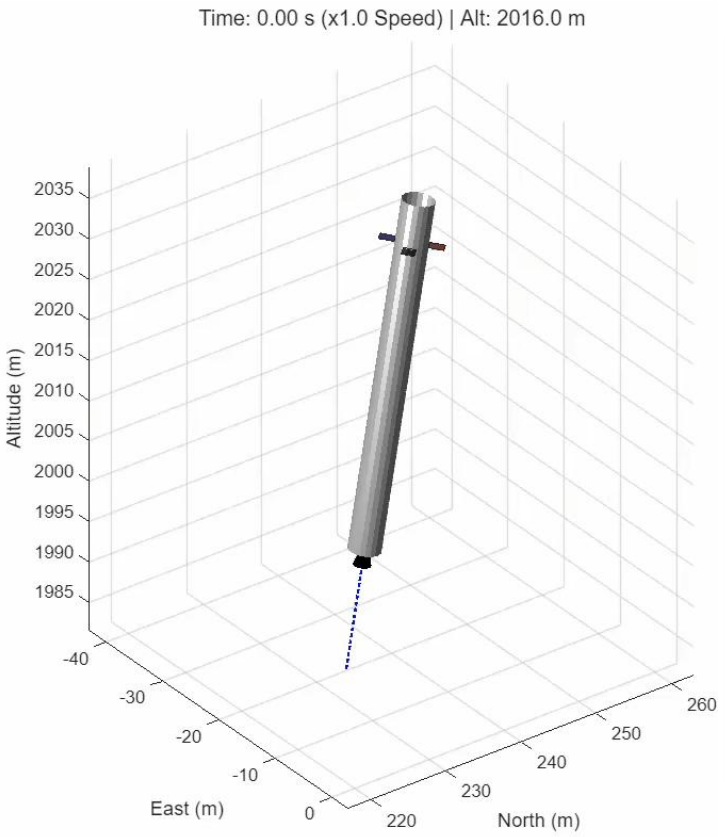
$$J = (\mathbf{Bu} - \boldsymbol{\tau}_{cmd})^T \mathbf{W}_\tau (\mathbf{Bu} - \boldsymbol{\tau}_{cmd}) + \mathbf{u}^T \mathbf{W}_u \mathbf{u}$$

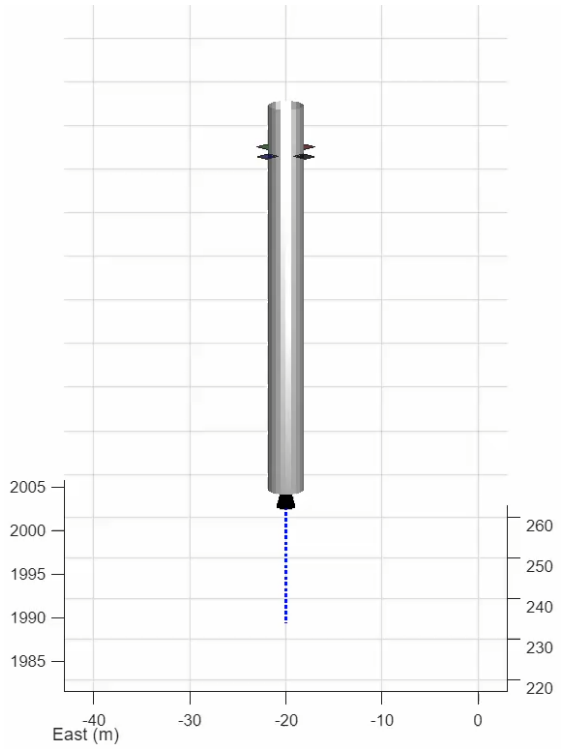
$$\mathbf{W}_\tau \in \mathbb{R}^{3\times 3}$$

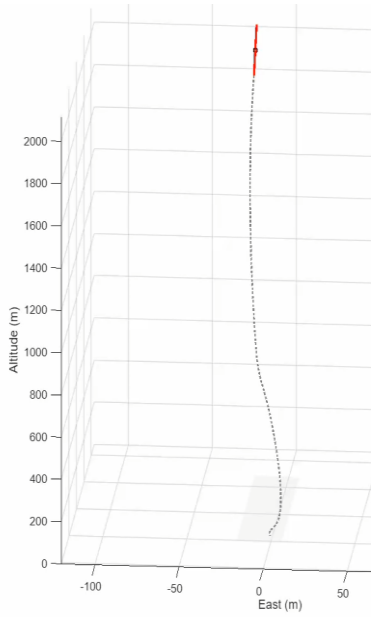
$$\mathbf{W}_u = \mathrm{diag}(\underbrace{1, 1}_{\text{TVC}}, \underbrace{W_{gf}, W_{gf}, W_{gf}, W_{gf}}_{\text{Grid Fins}}, \underbrace{W_{rcs}, W_{rcs}, W_{rcs}, W_{rcs}}_{\text{RCS}})$$

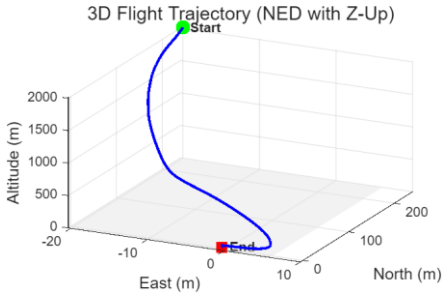


Simulation Results – PDG





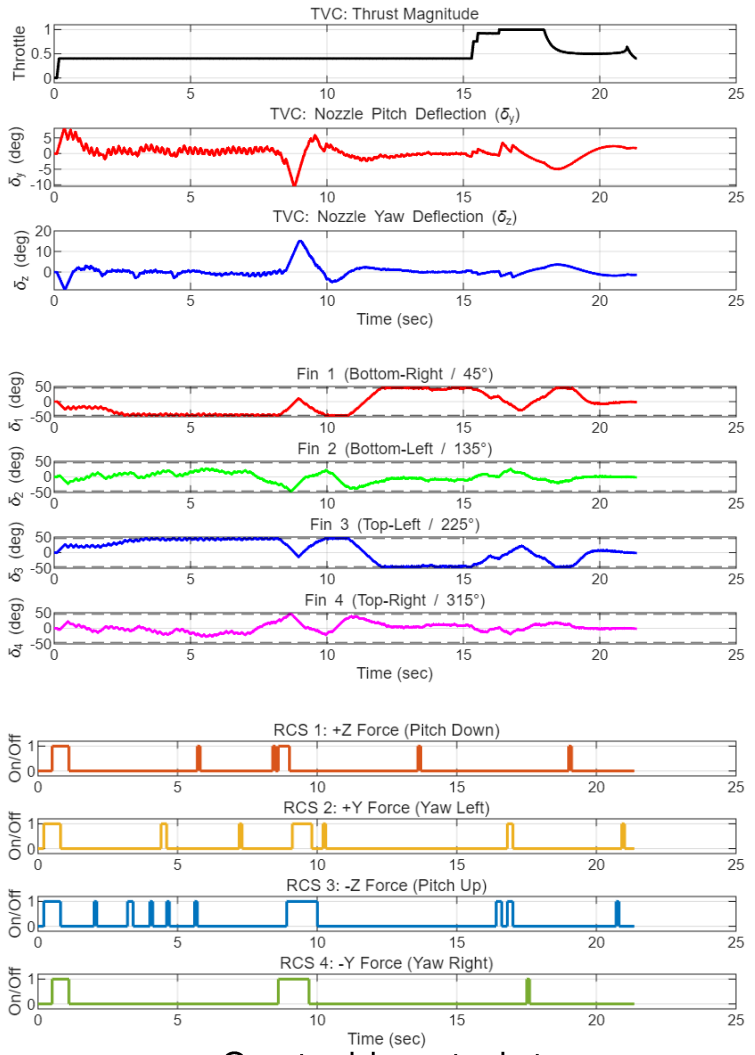




$r_{0,NED} = [240, -20, -2000\ ]\ m$

$v_{0,body} = [-184, 0, 0]\ m/s$

Landing point: $N = -0.3\ m, E - 0.5\ m$



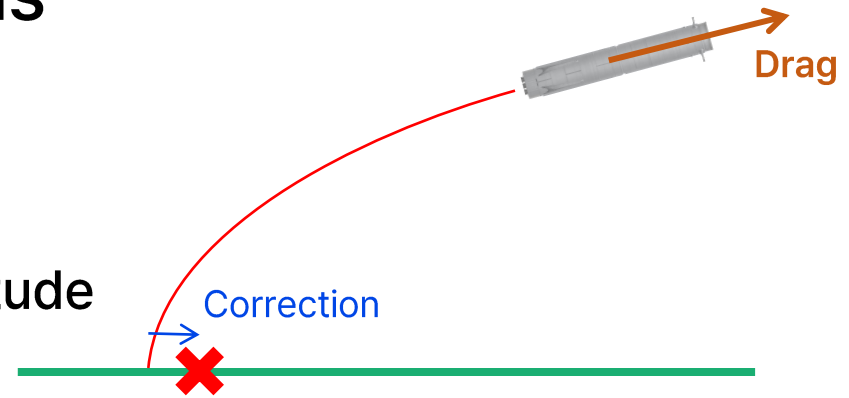
Control Input plot

TVC(top), Grid fin(middle), RCS(bottom)

SMRL Grid-fin Research

Grid-fin Null-space

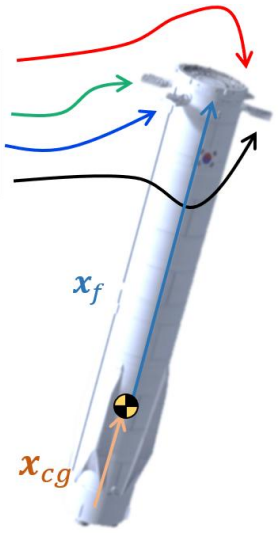
- Null-space drag modulation with grid Fins
  - 4 grid fins track a 3-axis torque command
  - This leaves 1 redundant DoF
  - That direction is the null space of B
  - Moving fins along it adds drag without changing attitude
  - Usable for downrange control instead of AoA

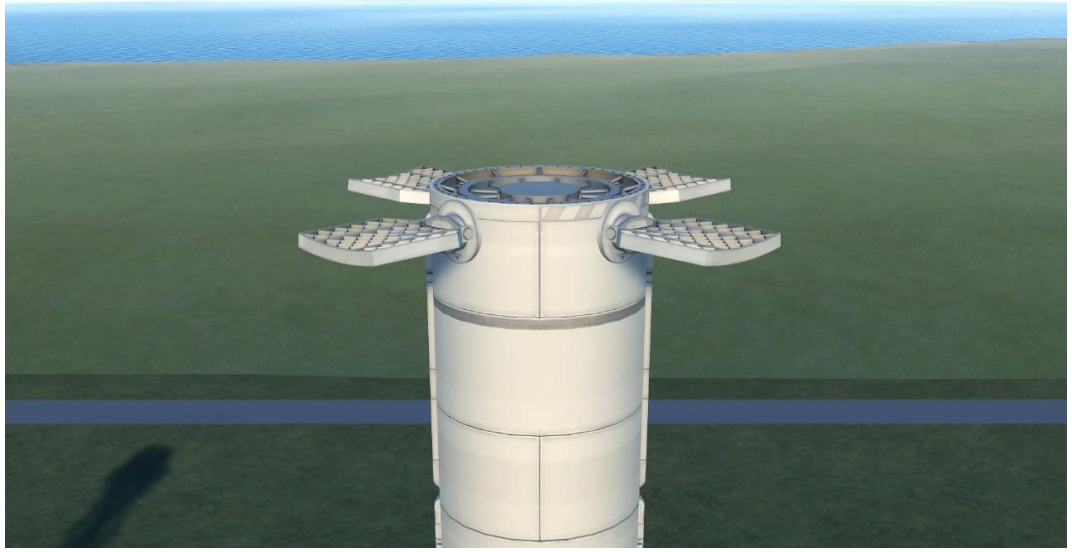


$$\boldsymbol{\tau}_{cmd} = \begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} \quad \boldsymbol{u}_p = B^{+}\boldsymbol{\tau}_{cmd} \quad \boldsymbol{u}_{fin} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}$$

$$u = u_p + N(u_p)\,\eta$$

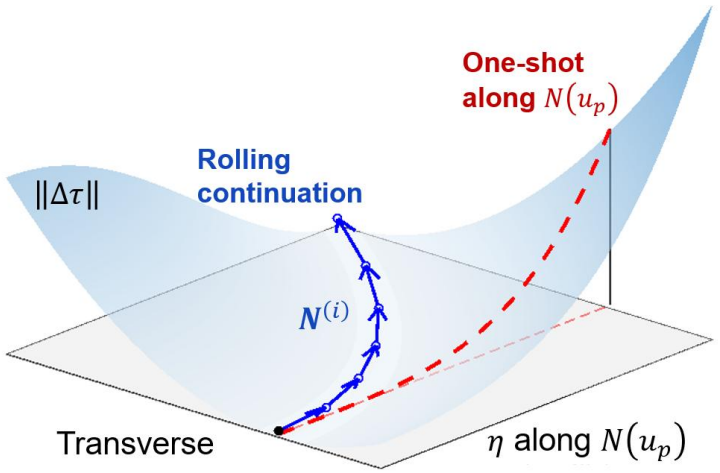
Null-space drag input



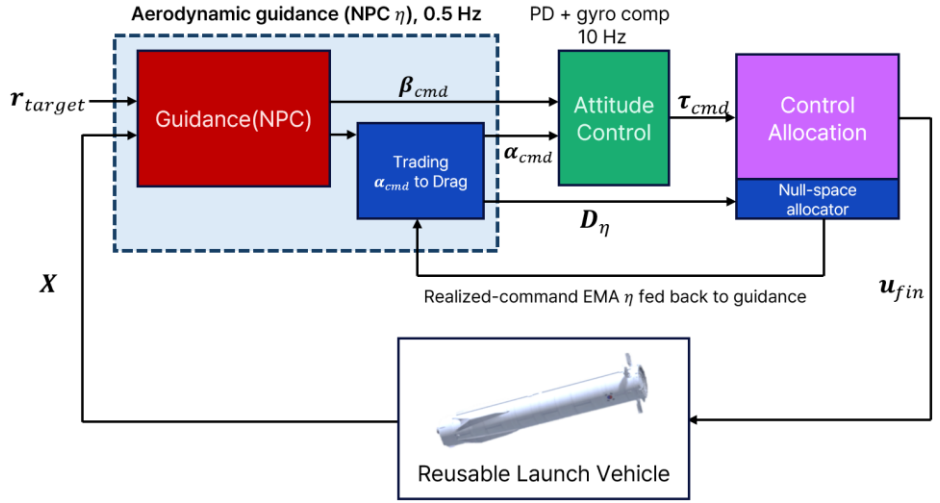


Grid-fin Null-space

- Rolling continuation & NPC-η architecture
  - Allocator walks the null direction
  - One pseudo-inverse fix: 6,000× less torque leak
  - NPC-η trades AoA command for drag



Rolling continuation vs one-shot step



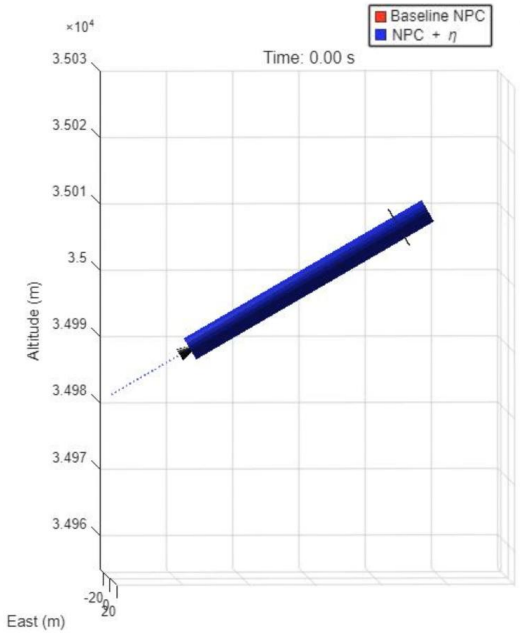
NPC-η guidance and allocation loop

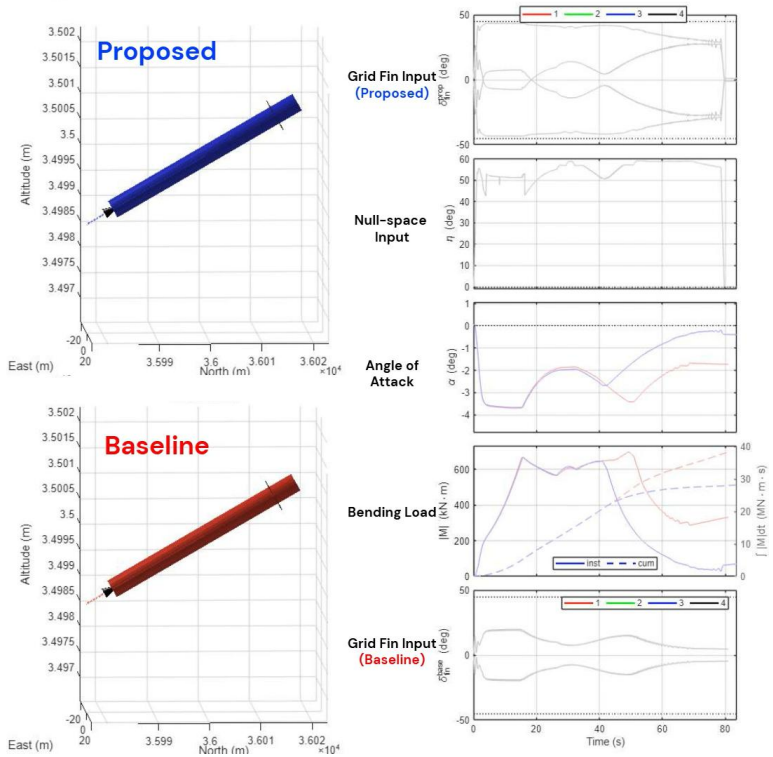
$$\mathbf{u} = \mathbf{u}_p + \mathbf{N}\eta, \qquad \mathbf{u}^{(j+1)} = \Pi_U\left(\mathbf{u}^{(j)} + \mathbf{N}^{(j)}\Delta\eta\right)$$
$$\mathbf{N}^{(j)} = \mathcal{N}\left(\mathbf{B}_M\left(\mathbf{u}^{(j)}\right)\right) \quad (no\ net\ torque)$$

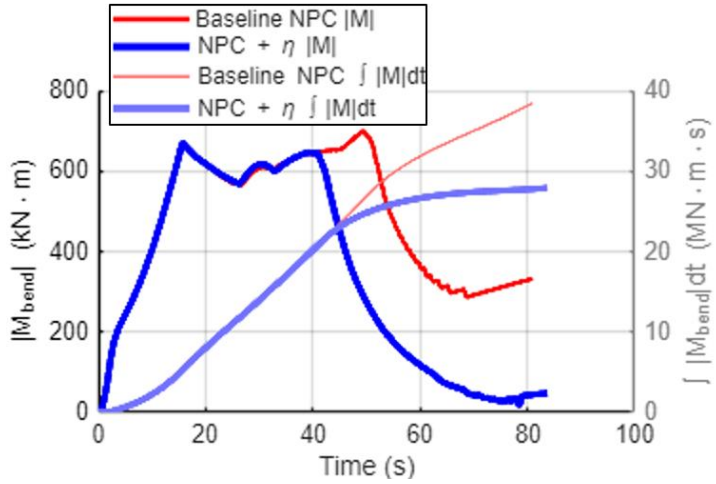
$$\alpha_{\mathrm{cmd}} = \alpha_0 - J_D/J_\alpha\, D_{\eta,\mathrm{cmd}}$$
$$D_\eta(t) = \lambda e^{-t/\tau_D} D_{\eta,\max}(M,\gamma)$$

Grid-fin Null-space

- Closed-loop results vs NPC baseline
  - Null-space drag takes over downrange control, so AoA drops
  - Cumulative bending −27% (38.4 → 27.9 MN·m·s)







Conclusion

- Summary
  - Grid-fin null space: a drag channel with no net torque
  - Rolling continuation: 6,000× less torque leak
  - NPC-η swaps AoA for drag: 27% less bending

- Future work
  - Monte Carlo with aero and wind uncertainty
  - Beam model for bending; bending-aware optimal guidance

Part 2

Speaker Introduction



ABOUT ME

Kangmuk Jang

Master student

Email: jangk4@snu.ac.kr

Space Mobility & Robotics Laboratory

(Advisor: Dr. Hyeongjun Park)

Research interest:

Robust Multi-Phase Descent Guidance for a Reusable Launch Vehicle

- Conference proceedings
  - **K. Jang**, J. Kim, and H. Park, “Robust Multi-Phase Descent Guidance for a Reusable Launch Vehicle,” 77th International Astronautical Congress (IAC 2026), Antalya, Türkiye, Oct. 2026
  - J. Kim, **K. Jang**, and H. Park, “Null-Space Drag Modulation with Grid Fins: A Load-Decoupled Downrange Control Channel for Reusable Launch Vehicle Aerodynamic Descent,” 77th International Astronautical Congress (IAC 2026), Antalya, Türkiye, Oct. 2026
  - **장강묵**, 김준석, 박형준, “재사용 발사체 다중 단계 하강 유도 및 전환 최적화,” 한국항공우주학회 춘계학술대회, 제주, 2026년 4월.
  - 김준석, **장강묵**, 박형준, “강화학습 기반 적응형 가중치 튜닝을 이용한 재사용 발사체 최적제어 연구,” 한국항공우주학회 춘계학술대회, 제주, 2026년 4월.
  - **장강묵**, 박형준, “궤도수송선을 이용한 다중 위성 배치 및 연료 최적 궤적 설계,” 한국항공우주학회 추계학술대회, 고성, 2025년 11월.

Guidance Model and Problem Formulation

Guidance Inputs and Outputs

- Guidance computes reference commands
  - Inputs: estimated state, target and vehicle limits
  - Outputs: state histories, attitude, and thrust references
- Guidance updates more slowly than control
  - Ex) guidance (5 Hz), control (50 Hz), dynamics / IMU (200 Hz)
  - These are design update rates, not measured execution times



Guidance Methods

- Common approaches to landing guidance

Method	Solution approach	Objective and constraints
ZEM / ZEV *	Analytical thrust command from terminal errors	Terminal state constraints No explicit path constraints
Direct NLP / SQP **	Solve the nonlinear trajectory problem	Fuel objective and explicit flight constraints
SCP / SCvx **	Solve a sequence of convex subproblems	Fuel objective and explicit flight constraints

- Why SCP for constrained landing guidance
  - Optimize fuel consumption subject to vehicle and flight constraints
  - Convex subproblems support efficient numerical solves
  - Outer iterations and nonlinear feasibility checks still take time

* Simplicio, P., et al., Journal of Spacecraft and Rockets, Vol. 57, No. 1, 2020, pp. 74–89, Sec. III.B, Eqs. (47)–(49)
** Malyuta, D., et al., IEEE Control Systems, Vol. 42, No. 5, 2022, pp. 40–113

Update Periods and Runtimes

- Guidance timing depends on the model and implementation

Study / method	Update period	Reported runtime	Implementation / scope
Simplicio (2020) * ZEM / ZEV	0.1 s (10 Hz) Simulator rate	Not quantified Analytical CTV	Analytical CTV law
KARI (2026) ** 3-DoF SCP, 30 nodes	3-DoF: 1 s 6-DoF: 2 s	0.9991 to 0.6964 s 3-DoF, five iterations	Static storage, direct CSC Inverse-free FOH Optimized derivatives
Wang (2019) *** 3-DoF MPC	Variable interval ≈ 0.4 s	≈ 0.4 s	Adaptive grid, parallel PISC C++ / MOSEK
Szmuk (2020) **** 6-DoF SCP	Not specified	Max. solver < 0.7 s	MATLAB / ECOS

  - CSC: compressed sparse column, FOH: first-order hold

* Simplicio, P., et al., Journal of Spacecraft and Rockets, Vol. 57, No. 1, 2020, pp. 74–89, Sec. III.B
** Kim, Y., KARI, KSAS Spring Conference, 2026
*** Wang, J., et al., Journal of Guidance, Control, and Dynamics, Vol. 42, No. 5, 2019, pp. 1078–1092, Sec. V.B
**** Szmuk, M., et al., Journal of Guidance, Control, and Dynamics, Vol. 43, No. 8, 2020, pp. 1399–1413, Sec. IV.D

Guidance Model and EOMs

- Forces in the guidance frame
  - Gravity and body-frame aerodynamics *

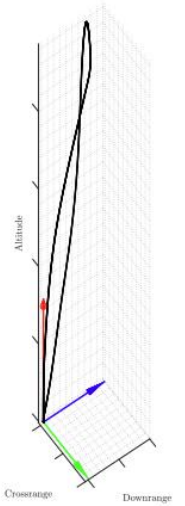
$$a_{grav} = -\mu_{\oplus} \frac{r + r_{T,DCA}}{\left\| r + r_{T,DCA} \right\|_2^3}$$

$$a_{aero}^{DCA} = \frac{1}{m} R_{BODY}^{DCA} F_{aero}^{BODY}(\alpha_1, \alpha_2, M, q_{dyn})$$

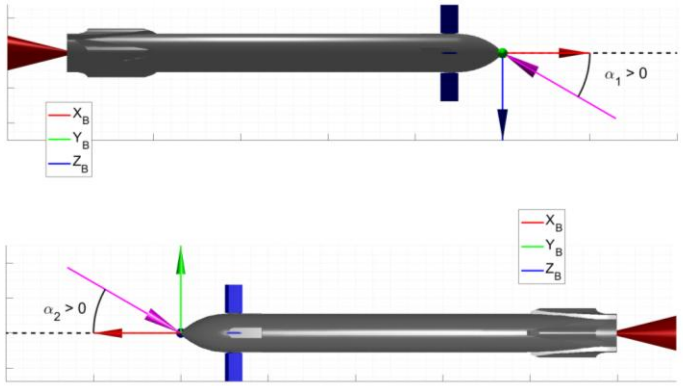
- Add the Earth-rotation terms
  - Target-centered DCA frame, fixed roll and engine off *

$$\dot{r} = v \quad \dot{\theta} = u_\theta \quad \dot{\psi} = u_\psi$$

$$\dot{v} = a_{grav} + a_{aero} - 2\omega \times v - \omega \times (\omega \times r)$$



DCA: Downrange-Crossrange-Altitude



Aerodynamic angles *

* Sagliano, M., et al., AIAA Scitech 2021 Forum, AIAA Paper 2021-0862, 2021

Powered Landing Dynamics

- Substitute thrust, gravity and aerodynamics
  - Thrust vector and acceleration terms are expressed in DCA

$$\dot{r} = v \quad x = (r^T, v^T, m, \varphi, \theta, \psi, T_{atmo})^T$$

$$\dot{v} = \frac{T_{thr} + R_{BODY}^{DCA} F_{aero}^{BODY}}{m} - \mu_{\oplus} \frac{r + r_{T,DCA}}{\|r + r_{T,DCA}\|_2^3} - 2\omega \times v - \omega \times (\omega \times r)$$

- Extend the state with mass and thrust
  - Controls are Euler-angle rates and atmospheric thrust rate
  - Nozzle area and pressure relate vacuum to atmospheric thrust

$$\dot{m} = -\frac{T_{vac}}{I_{sp}\, g_0} \quad T_{atmo} = T_{vac} - A_{nz}\, p$$

$$\dot{\varphi} = u_{\varphi} \quad \dot{\theta} = u_{\theta} \quad \dot{\psi} = u_{\psi} \quad \dot{T}_{atmo} = u_T$$

Landing Objectives and Constraints

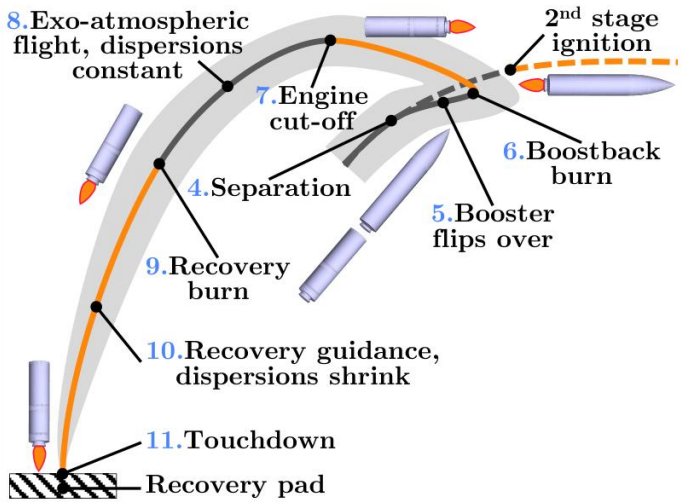
- Terminal landing state
  - Initial state and terminal position/velocity

$$x(t_0) = x_0 \quad r(t_f) = r_f \quad v(t_f) = v_f$$

- Constrain the flight envelope
  - Dynamic pressure, load and glideslope limits
  - Bounds on states and control rates

$$q_{dyn} = \frac{1}{2}\rho V^2 \leq q_{max} \quad n_z = \frac{\|F_n\|}{m\, g_0} \leq n_{z,max}$$

$$\frac{r_A}{\|r_{D,C}\|} \geq \cot \gamma_{gs} \quad x_L \leq x \leq x_U \quad u_L \leq u \leq u_U$$



RTLS (Return to Launch Site) mission

Illustrative design limits

$q_{max} = 65kPa$

$n_{z,max} = 5$

$\gamma_{gs} = 75°$ (measured from local vertical)

Trajectory Optimization and Online SCP Simulation

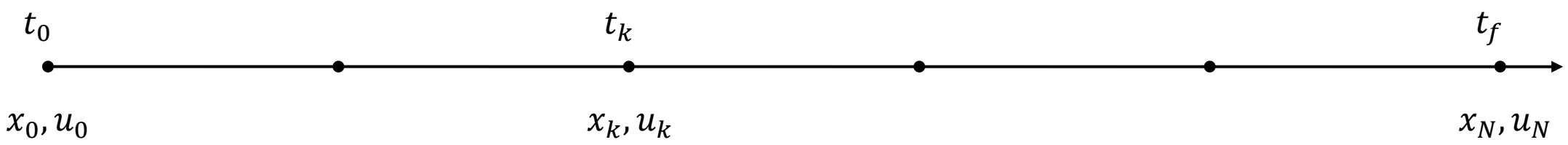
Trajectory Optimization Problem

- Optimize the trajectory and flight duration
  - Minimize propellant use and penalize control rates

$$min\, J \;=\; -w_m\, m(t_f) \;+\; w_u \int_{t_0}^{t_f} u^T\, W_u\, u\, dt$$

- Enforce the model and flight constraints
  - Use the dynamics, terminal conditions, and path limits as defined previously
  - Optimize state and input values at discrete time nodes

$$\dot{x} \;=\; f(x,u) \quad x(t_0) \;=\; x_0 \quad x(t_f) \;\in\; X_f$$



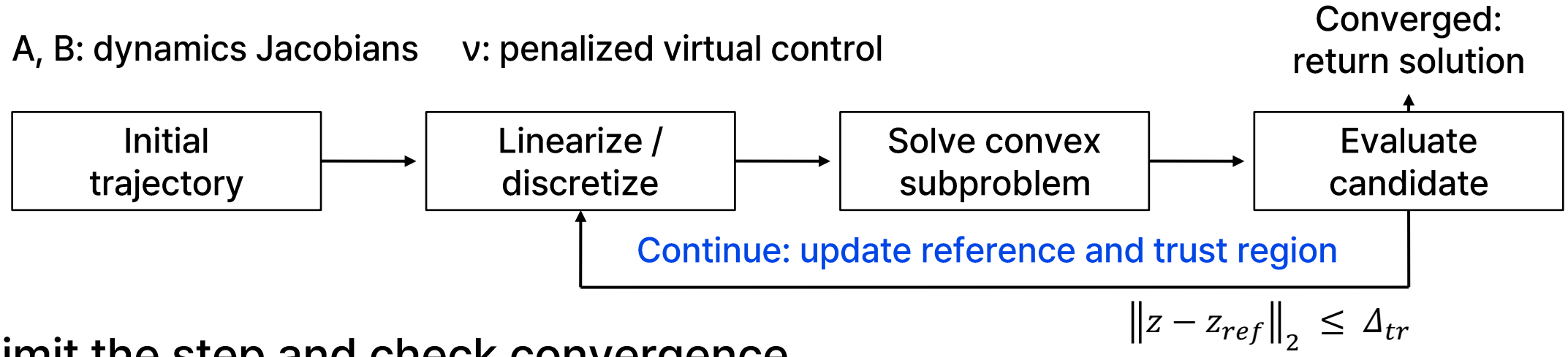
Interpolate the inputs between nodes

SCP Solution Process

- Solve convex subproblems around a reference trajectory
  - At iteration j, linearize the model around the current reference

$$\dot{x} \approx f\left(x^{(j)}, u^{(j)}\right) + A^{(j)}\left(x - x^{(j)}\right) + B^{(j)}\left(u - u^{(j)}\right) + \nu$$

A, B: dynamics Jacobians ν: penalized virtual control



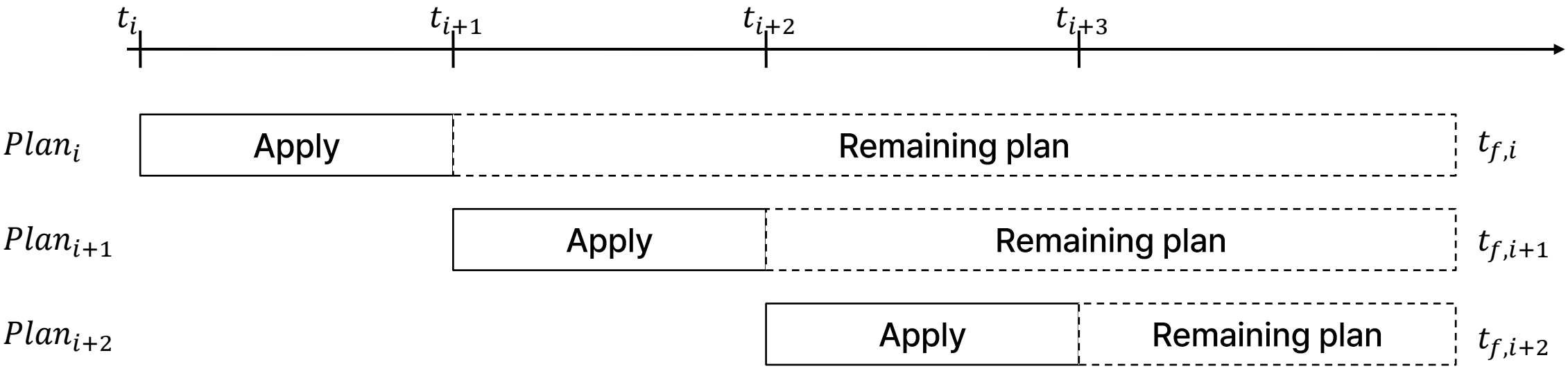
- Limit the step and check convergence
  - Trust region limits the step from the reference trajectory
  - Penalized virtual control relaxes the linearized dynamics
  - Check cost change, nonlinear residuals, and virtual control before stopping

Online Trajectory Updates

- Update the remaining trajectory from the latest state estimate
  - Apply the next reference segment, then update again

$x_{pred,i}(t_i) = \hat{x}(t_i)$

i: guidance update



  - Use the previous solution as the next initial guess

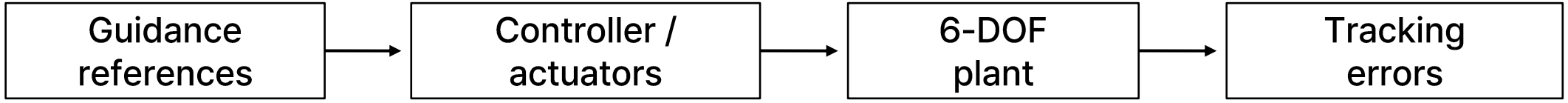
Guidance Validation in Simulation

- Check the trajectory in the nonlinear guidance model
  - Integrate optimized inputs and check terminal and path residuals



$$e_{sim}(t) = x_{sim}(t) - x^{*}_{pred,i}(t)$$

- Evaluate tracking in the 6-DOF closed loop
  - Include the controller, actuators, model mismatch, and disturbances
  - Check landing error, fuel use, constraint violations, and update timing

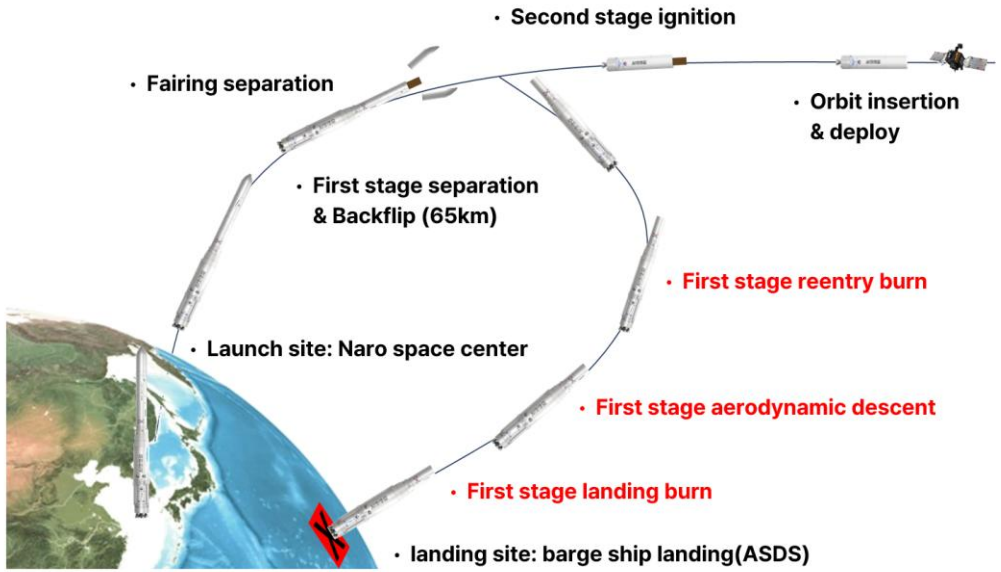


$$e_r = r_{6DOF} - r_{ref} \quad e_v = v_{6DOF} - v_{ref}$$

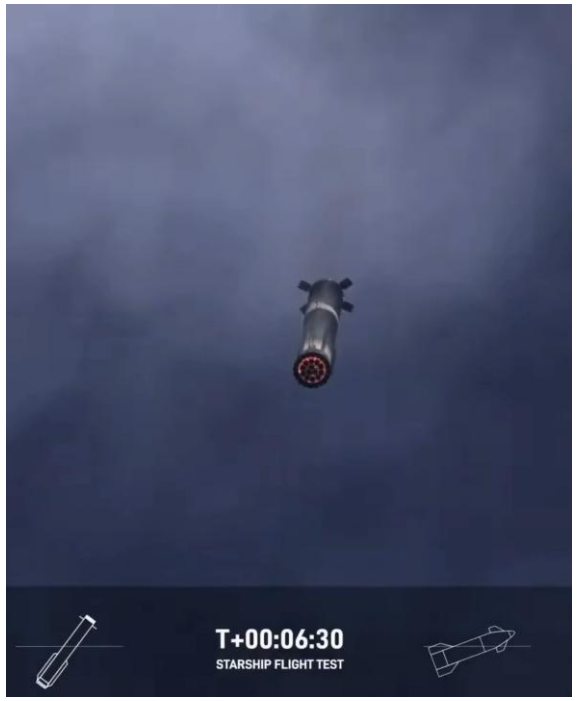
SMRL RLV Guidance Research

Motivation

- RLV guidance involves distinct dynamics and critical phase transitions
  - Distinct flight regimes: reentry, aerodynamic descent, and precise landing phases
  - Highly nonlinear and phase-dependent dynamics
  - Robust and feasible trajectory guidance is critical



RLV Concept of operations

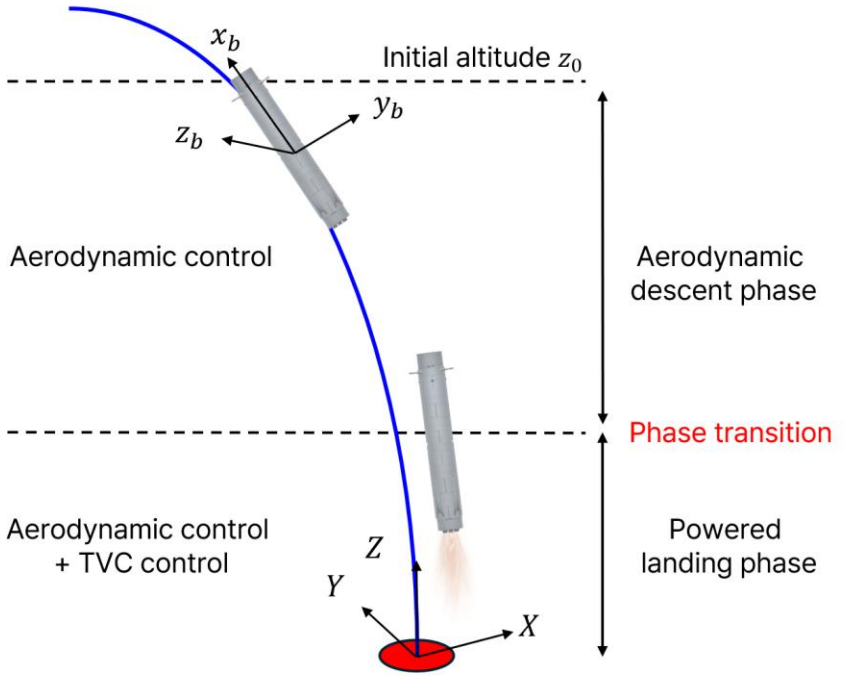


Starship phase transition in landing guidance *

* Rocketry, "How Grid Fins Steer a 440,000-Pound Super Heavy," YouTube video, URL: https://www.youtube.com/watch?v=crWNxruBQzA

Vehicle Dynamics

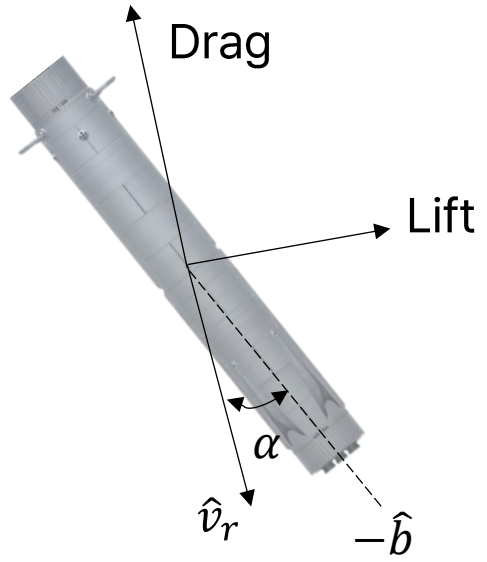
- Coordinate frames
  - Local landing frame (X, Y, Z) for point mass 3-DoF translational motion
  - Body-axis orientation defines aerodynamic and thrust directions
  - Pitch and yaw use a local vertical reference frame
- State vector
  - $x = [r^T, v^T, m, \theta, \psi, T]^T$
  - r^T, v^T: position and velocity in local frame
  - m, T: mass and thrust magnitude
  - θ, ψ: pitch and yaw angles
- Control input
  - $u = \left[u_\theta, u_\psi, u_T\right]^T$
  - u_θ, u_ψ, u_T: pitch/yaw rate and thrust rate commands



Multi-phase descent guidance diagram

Aerodynamic Descent EOMs

- Lift and drag-based trajectory shaping
  - $\dot{\theta}$ (pitch rate) and $\dot{\psi}$ (yaw rate) change body-axis orientation
  - Attitude-dependent aerodynamic forces shape the descent trajectory
  - Mach-dependent reference coefficients are used



$$u_\theta = \dot{\theta}$$
$$u_\psi = \dot{\psi}$$

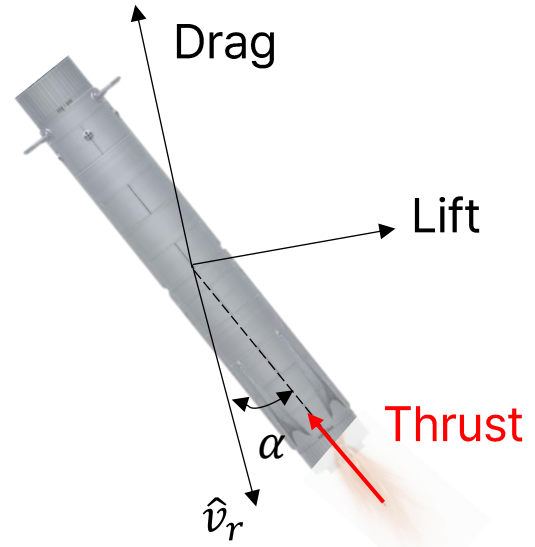
$$\hat{b} = \begin{bmatrix} \sin\theta \\ \cos\theta \sin\psi \\ \cos\theta \cos\psi \end{bmatrix}$$
$$\alpha = \cos^{-1}(-\hat{b} \cdot \hat{v}_r)$$
$$\hat{L} = \frac{-\hat{b} + (\hat{b}^T \hat{v}_r)\hat{v}_r}{\left\| -\hat{b} + (\hat{b}\, \hat{v}_r)\hat{v}_r \right\|}$$

$$C_L = K_L(M) \sin\alpha$$
$$C_D = C_{D0}(M) + K_D(M) \sin^2\alpha$$
$$a_{lift} = \frac{\bar{q} S_{ref}}{m} C_L \hat{L}$$
$$a_{drag} = -\frac{\bar{q} S_{ref}}{m} C_D \hat{v}_r$$

$$\dot{r} = v$$
$$\dot{v} = g + a_{lift} + a_{drag}$$

Powered Landing EOMs

- Thrust vector control (TVC)
  - Body attitude (θ, ψ) determines the thrust direction along the body axis
  - Thrust magnitude (T) determines thrust acceleration, with ideal tracking of attitude and thrust rate commands assumed



$$u = \left[u_\theta, u_\psi, u_T\right]^T = \left[\dot{\theta}, \dot{\psi}, \dot{T}\right]^T$$

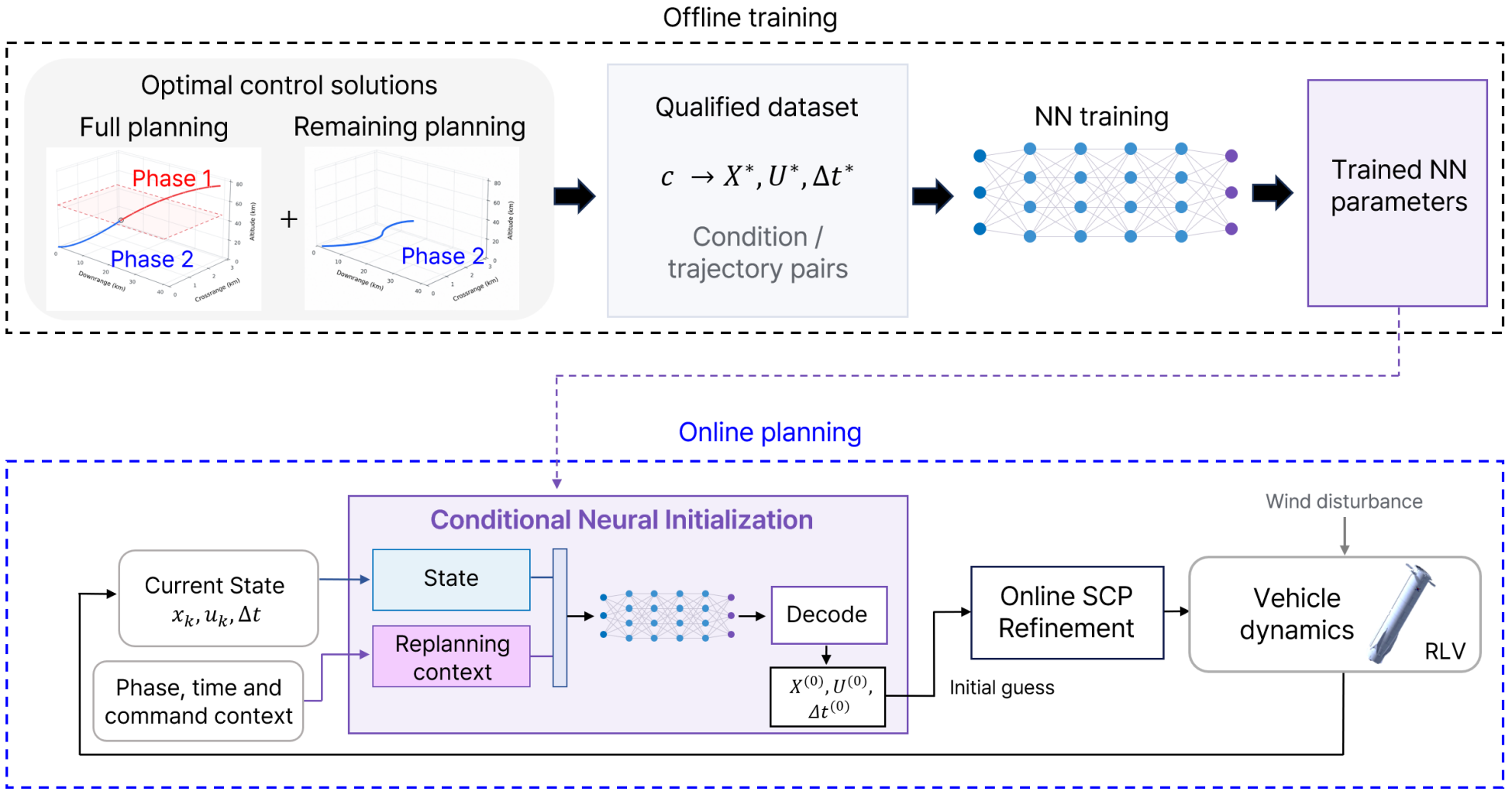
$$a_{Thrust} = \left(\frac{T}{m}\right)\begin{bmatrix} \sin\theta \\ \cos\theta \sin\psi \\ \cos\theta \cos\psi \end{bmatrix}$$

$$\dot{r} = v$$

$$\dot{v} = g + a_{lift} + a_{drag} + a_{Thrust}$$

$$\dot{m} = -\frac{\mathrm{T}}{I_{sp} g_0}$$

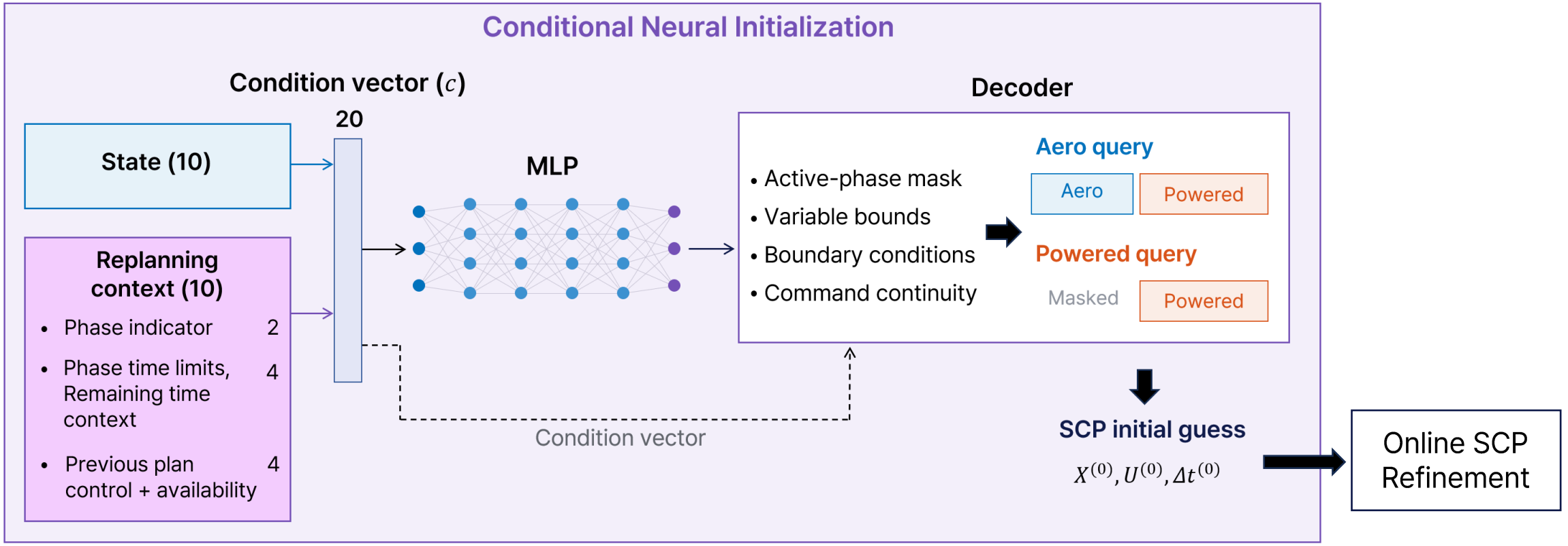
CNI-SCP Architecture



Algorithm Flowchart

Conditional Neural Initialization

- State and replanning conditions define the remaining trajectory
  - MLP predicts state, control, and phase-duration initial guesses
  - Decoder uses the same condition vector to assemble remaining phases



CNI architecture

Data Generation

- Multi-phase trajectory optimization
  - Full and remaining-horizon problems generate reference trajectories
  - Qualified solutions form training pairs of conditions and trajectories

Multi-phase optimal control problem

$$J = -m(t_f) + \int_0^{t_s} \left(\|u_{ph_1}\|_{W_u}^2 \right) dt + \int_{t_s}^{t_f} \left(\|u_{ph_2}\|_{W_u}^2 \right) dt + J_{smooth}$$

$$\begin{bmatrix} -u_{\theta_{max}} \\ -u_{\psi_{max}} \end{bmatrix} \leq \begin{bmatrix} u_\theta \\ u_\psi \end{bmatrix} \leq \begin{bmatrix} u_{\theta_{max}} \\ u_{\psi_{max}} \end{bmatrix}$$

$$\bar{q} \leq \bar{q}_{max} \qquad \alpha \leq \alpha_{max}$$

$$T_{min} \leq T \leq T_{max} \qquad \cos\gamma_{max} \leq \hat{b}^T e_z$$

Phase linkage and boundary conditions

$$t_f = t_{ph_1} + t_{ph_2}$$

$$[r^T, v^T, m, \theta, \psi]^T(t_s^-) = [r^T, v^T, m, \theta, \psi]^T(t_s^+)$$

$$[u_\theta, u_\psi]^T(t_s^-) = [u_\theta, u_\psi]^T(t_s^+)$$

$$T(t_s^-) = 0 \qquad T(t_s^+) = T_{min}$$

$$x(0) = x_0$$

$$r(t_f) = r_f \qquad v(t_f) = v_f$$

$$\theta(t_f) = \psi(t_f) = 0$$

Qualified solutions ➡ Training dataset

Network Training

- Network learns using state and replanning context
  - Network predicts states, controls, and phase durations
  - Training combines reference matching, dynamics consistency, and path constraint penalties

$\rho(\cdot)$: Huber loss
$g(\cdot)$: Path constraint residuals
f_{ph}: Discrete-time dynamics
$ph \in \{ph_1, ph_2\}$

$$\min_{\eta} \mathcal{L} = \mathcal{L}_{supervised} + \mathcal{L}_{dynamics} + \mathcal{L}_{path}$$

Supervised trajectory loss

$$\mathcal{L}_{supervised} = \lambda_{state}\rho\left(W_x(\hat{x}_k - x_k^*)\right) + \|\hat{u}_k - u_k^*\|^2 + \left(\frac{(\hat{t}_{ph} - t_{ph}^*)}{t_{ph}^*}\right)^2$$

Dynamics consistency loss

$$\mathcal{L}_{dynamics} = \lambda_{dynamics}\rho\left(W_{dynamics}e_k\right) \qquad e_k = \hat{x}_{k+1} - f_{ph}\left(\hat{x}_k, \hat{u}_k, \hat{u}_{k+1}, \frac{\hat{t}_{ph}}{N_{ph}}\right)$$

Path penalty loss

$$\mathcal{L}_{path} = \lambda_{path}\rho(\max(0, g(\hat{x}_k)))$$

Simulation Setup

- Two-phase landing guidance with wind disturbance
  - Scenario 1: 10 m/s wind
  - Scenario 2: 20 m/s wind

Simulation parameters

Parameter	Value
Vehicle initial mass [kg]	55,000
Specific impulse [s]	443
Min / max thrust [kN]	412.68 / 1,375.6
Initial position [m]	[1,500 ± 100, 1,000 ± 100, 10,000]
Initial velocity [m/s]	[-30 ± 5, -10 ± 5, -250 ± 10]
Initial pitch / yaw [°]	±1
Replanning interval [s] aero / powered / altitude ≤ 100 m	3 / 2 / 1
Wind magnitude [m/s] Direction: 0°-360°	10, 20
Offline data generation	MATLAB fmincon (SQP)
Online convex solver	MOSEK 11.1

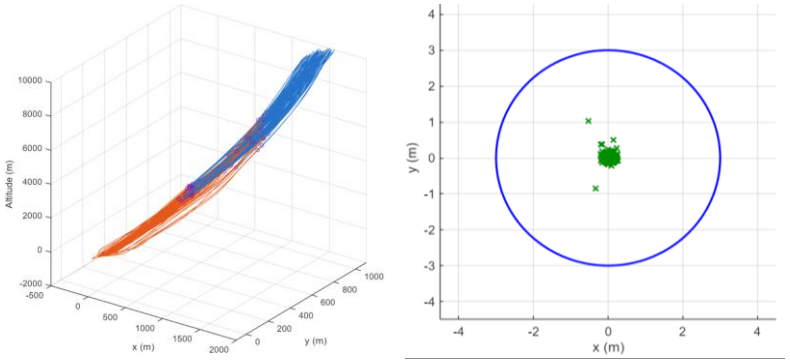
Training parameters

Parameter	Value
Training / validation	2,000 / 300 samples
Condition vector / reference trajectory	20 / 627 values
Hidden layers	4 x 512, SiLU
Trajectory grid	25 nodes per phase
Optimizer	AdamW
Epochs / batch size	300 / 128
Learning rate	0.0003
Weight decay	0.0001
Loss weights (λ_{state} / $\lambda_{dynamics}$ / λ_{path} / $\lambda_{\Delta u}$)	1 / 1 / 0.1 / 0.05

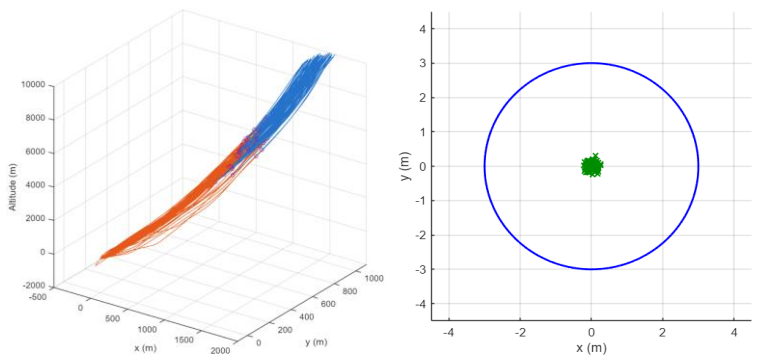
Simulation Result (1/3)

- Scenario 1: 10 m/s wind (100 Monte Carlo runs)

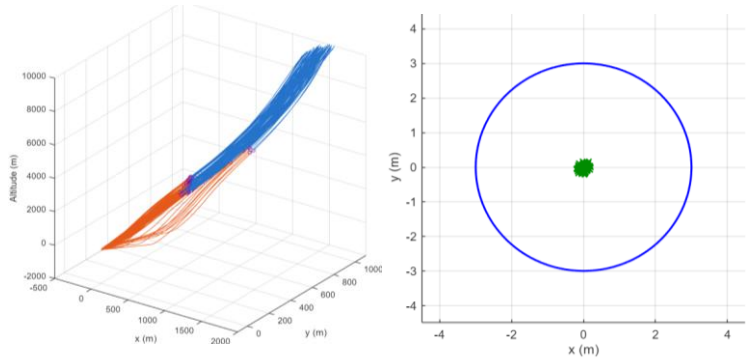
Phase 1
Phase 2



Cold-start SCP



Warm-start SCP



CNI-SCP (Proposed)

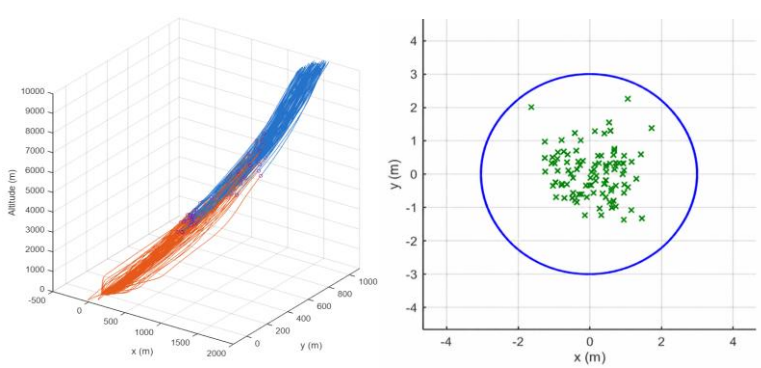
Guidance performance under 10 m/s wind

Method	Horizontal RMSE / Max [m]	SCP wall time: Mean / P95 [s]	Mean SCP iterations
Cold-start SCP	0.229 / 1.155	0.891 / 2.571	22.76
Warm-start SCP	0.159 / 0.332	0.836 / 2.539	12.92
CNI-SCP (Proposed)	**0.140 / 0.240**	**0.717 / 1.527**	**10.2**

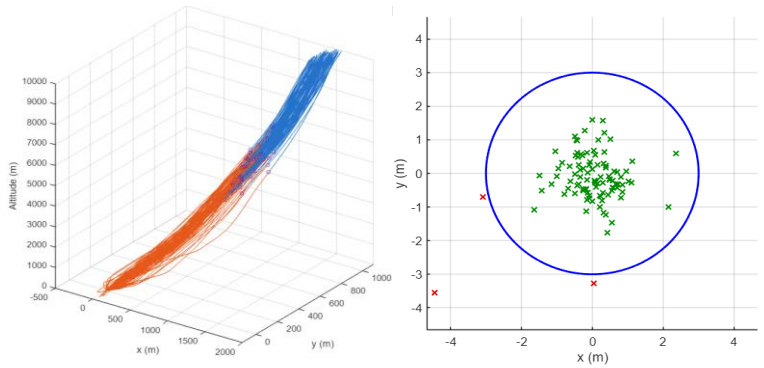
Simulation Result (2/3)

- Scenario 2: 20 m/s wind (100 Monte Carlo runs)

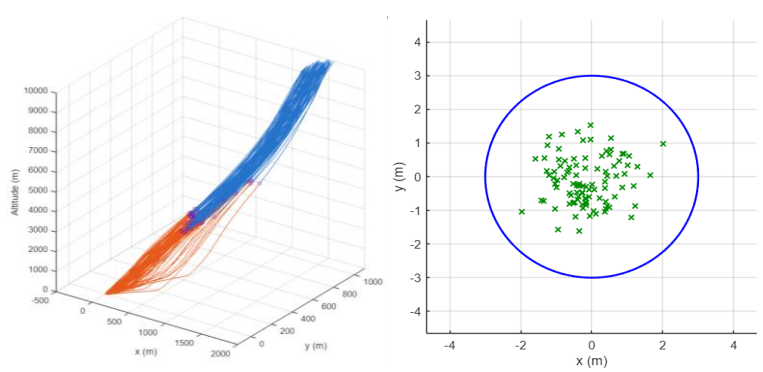
Phase 1
Phase 2



Cold-start SCP



Warm-start SCP



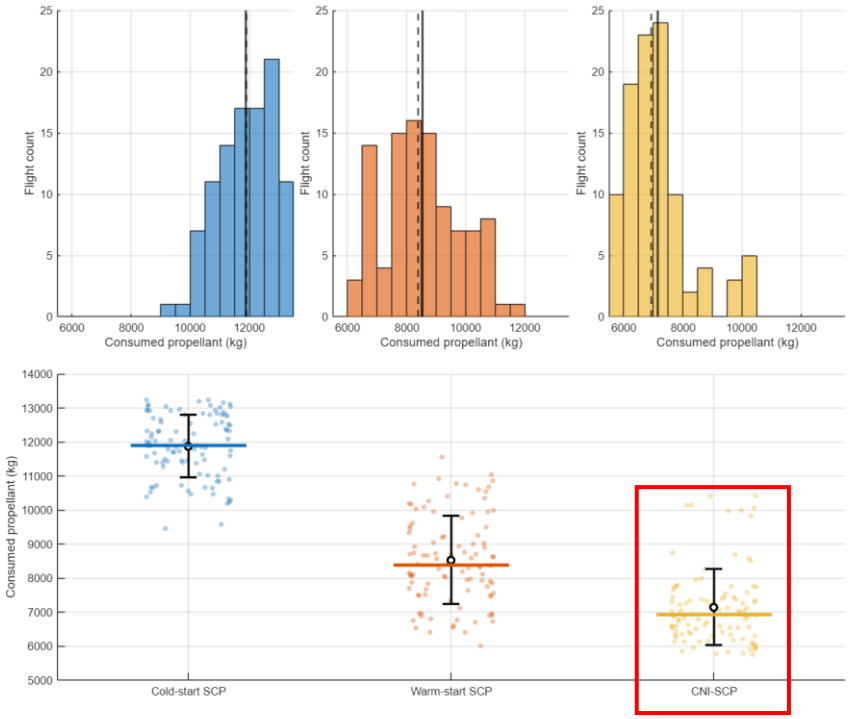
CNI-SCP (Proposed)

Guidance performance under 20 m/s wind

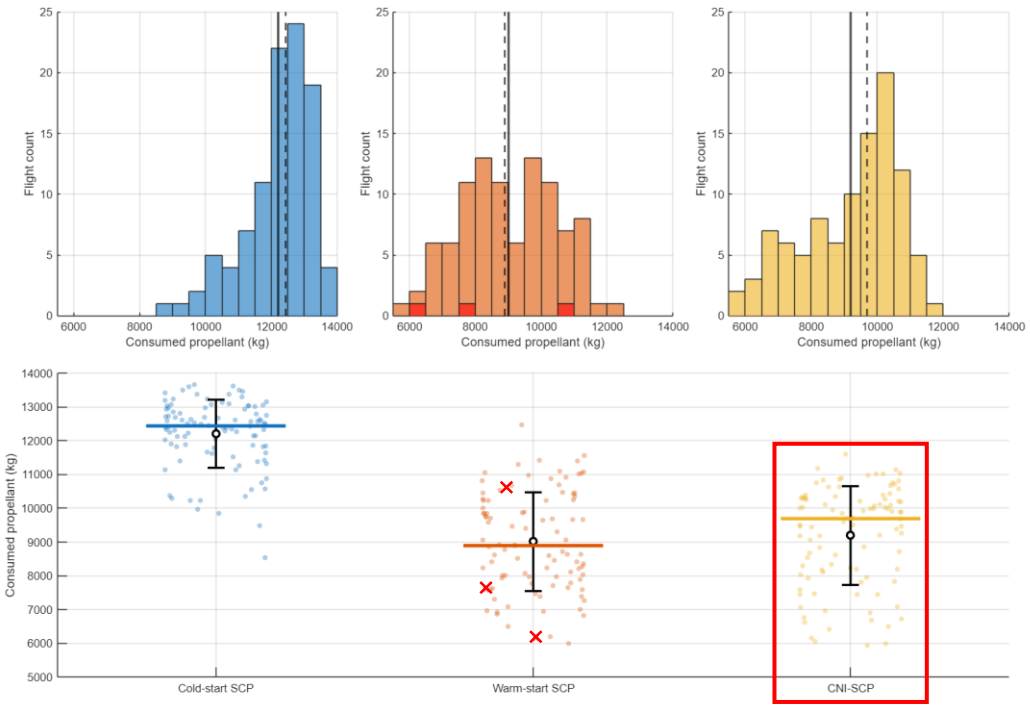
Method	Horizontal RMSE / Max [m]	SCP wall time: Mean / P95 [s]	Mean SCP iterations
Cold-start SCP	1.051 / 2.599	0.877 / 2.566	22.5
Warm-start SCP	1.164 / 5.683	0.860 / 2.559	15.56
CNI-SCP (Proposed)	**1.042 / 2.234**	**0.754 / 1.543**	**10.42**

Simulation Result (3/3)

- Propellant consumption comparison
  - At 10 m/s, CNI-SCP achieves the lowest mean propellant consumption
  - At 20 m/s, it uses 24.7% less propellant than Cold-start SCP and improves landing accuracy over Warm-start SCP with slightly higher consumption



Propellant consumption (10 m/s case)



Propellant consumption (20 m/s case)

Conclusion

- Summary
  - Proposed CNI-SCP for multi-phase guidance using conditional neural predictions to initialize online SCP
  - Achieved the lowest horizontal landing RMSE among three methods under 10 m/s and 20 m/s wind disturbances
  - Reduced SCP solve time and propellant consumption relative to Cold-start SCP
- Future work
  - Expand guidance range, including ballistic and reentry burn phases
  - Evaluate performance under broader flight conditions and aerodynamic uncertainties

Thank you